LinUCB Is (Almost) All You Need: A Regime Analysis of LLM Warm-Starts in Contextual Bandits

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Abstract

A recent line of work reports striking "reward multiplication": combining a frozen LLM policy with an adaptive contextual bandit dominates either component, sometimes by an order of magnitude. We argue this is a *regime indicator*, not a generic phenomenon — "cold-start LinUCB is (almost) all you need." We isolate three structural axes — effective dimension versus horizon, **feature coherence**, and **action-gap structure** — and prove a "LinUCB-suffices" bound along each. Saturating axis A or axis B alone certifies that no warm-start can improve cold-start LinUCB by more than a constant factor; axis C provides a gap-dependent save when the effective dimension is moderate. The three-way separation behind the multiplication story lives where axes A and B jointly deviate; a regret-balancing analysis sharpens this to a phase transition at $r \asymp \sqrt{T}/\log T$. Four pre-runtime diagnostics classify any instance into one of four regimes before any algorithm is run, and on every configuration we test correctly predict the sign and order of magnitude of the warm-start lift.

1 Introduction

Sequential decision making studies how an agent adapts to an unknown environment through repeated interaction [55, 57], with applications in recommendation [31, 3], healthcare [47, 46], and dialogue [30]. Large pretrained models, especially large language models (LLMs), supply strong prior knowledge that can guide early decisions [11, 34, 2, 8, 51].

A striking observation is that combining a frozen LLM with an adaptive contextual bandit appears to strictly dominate either component in isolation, sometimes by orders of magnitude [12, 23]. The phenomenon, dubbed *reward multiplication*, is empirically robust and intuitive in mechanism. Yet bandit warm-starting [10, 52] and confidence-gated model selection [9, 6, 35] are not new objects, and the older literature does *not* report comparable multipliers. We take that discrepancy seriously and ask the dual question: *when is plain cold-start LinUCB enough?* If "almost always," multiplication is a regime indicator, and the question shifts from "how do we get it?" to "what structural property of the dataset enables it, and can we predict it in advance?"

Our headline claim is that cold-start LinUCB is competitive with any warm-started variant up to a constant factor on *any* instance saturating one of the three structural axes below; in our experiments, this covers every configuration except one. The classical multiplicative gain requires the joint failure of **(A) ambient versus effective dimension** and **(B) feature coherence**, with **(C) action-gap structure** acting as a third (gap-dependent) save only when the effective dimension is moderate. The three-way separation underlying the original mechanism story ([12], restated as Proposition 1) lives at the corner where axes A and B both fail; axis C is saturated in that construction (uniform

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

gap 1) yet warm-starting still helps because the gap-dependent bound of Lemma 3 is vacuous when $r = T$. The axes are sufficient conditions in priority order: axis A is the dimensional phase transition $r \asymp \sqrt{T}/\log T$ (primary); axis B saturates whenever features have positive coherence (typical of pretrained embeddings); axis C saves only when r is moderate enough for the gap-dependent bound to be non-vacuous. We render this as a hierarchical decision tree (Figure 1); reaching the bottom "warm-start-essential" leaf requires axes A and B both to fail and axis C either to fail or to be vacuous due to large r. Four dataset-level statistics classify any instance into one of four regimes before any algorithm is run, correctly predicting whether warm-starting helps on every configuration we test (Section 4), including previously unexplained negative lifts and the multi-fold gains uniquely flagged as warm-start-essential. We provide a more comprehensive related works in Appendix A.

Our contributions.

- **Regime-indicator framework.** A "LinUCB-suffices" lemma along each axis (Lemmas 1, 2, 3) paired with a sharp-threshold extension of the three-way separation in the complementary direction (Lemmas 6, 5, 7, in appendix). The original separation [12] (Proposition 1) is the joint corner (Corollary 1).
- **Confidence-gated meta-algorithm with regime-aware guarantees.** Theorem 1 gives shrinkage factor $\Psi(\tau, \nu, T) \approx \sqrt{\lambda/(\lambda + \nu\tau)}$, bounded strictly below 1 horizon-independently *only* when all three axes deviate. The phase transition $r \asymp \sqrt{T}/\log T$ (Theorem 2, Corollary 2) is the formal version of axis A.
- **Pre-runtime diagnostics, empirically validated.** Four measurable statistics classify each instance *before* any decision-making algorithm runs and correctly predict sign and order of magnitude of the warm-start lift on every configuration tested — including negative lifts not predicted by previous treatments.

2 Preliminaries

To formalize the question above, we work in a standard linear contextual bandit setting [55, 57]. At each round $t = 1, \ldots, T$, the learner observes a finite action set $\mathcal{A}_t$, where each $a \in \mathcal{A}_t$ has a feature vector $x_{t,a} \in \mathbb{R}^d$ with $\|x_{t,a}\|_2 \le L$. The learner selects an action $A_t \in \mathcal{A}_t$ and observes reward

$$r_t \;=\; \langle \theta^\star, x_{t,A_t} \rangle + \eta_t,$$

where $\theta^\star \in \mathbb{R}^d$ is unknown with $\|\theta^\star\|_2 \le S$ and η_t is conditionally mean-zero σ-sub-Gaussian noise. Write $\mu_t(a) = \langle \theta^\star, x_{t,a} \rangle$ for the expected reward and $a_t^\star \in \arg\max_a \mu_t(a)$ for the optimal action. The cumulative pseudo-regret of a policy $\pi = (\pi_1, \ldots, \pi_T)$ is $R_T(\pi) = \sum_t \big(\mu_t(a_t^\star) - \mu_t(\pi_t)\big)$. The learner has access to two qualitatively different decision rules:

1. a *frozen policy* induced by some estimator $\widehat{\theta} \in \mathbb{R}^d$ with associated design matrix $V \succ 0$, never updated online;
2. an *adaptive bandit algorithm* (LinUCB), which maintains $V_t = \lambda I + \sum_{s<t} x_{s,A_s} x_{s,A_s}^\top$, response $b_t = \sum_{s<t} r_s x_{s,A_s}$, parameter $\widehat{\theta}_t = V_t^{-1} b_t$, and selects

$$A_t \in \arg\max_a \langle \widehat{\theta}_t, x_{t,a} \rangle + \beta_t \|x_{t,a}\|_{V_t^{-1}}.$$

The frozen policy may be interpreted as an LLM, a regression head on pretrained features, or any other offline-trained predictor. The adaptive algorithm is the standard online learner. The structural asymmetry is that the frozen policy may be good initially but cannot improve, while the adaptive algorithm may be weak initially but improves with data. The question of this paper is: *when, if ever, is the frozen policy strictly necessary, in the sense that combining it with the adaptive algorithm strictly outperforms running the adaptive algorithm alone?*

Cold-started and warm-started LinUCB. Throughout the paper we compare two specific instantiations of the adaptive algorithm above, distinguished only by their initial state. *Cold-started LinUCB* is item 2 above with $V_1 = \lambda I_d$ and $b_1 = 0$. *Warm-started LinUCB* is the same algorithm with

$$V_1 \;=\; \lambda I_d + M, \qquad b_1 \;=\; c, \tag{1}$$

for some symmetric positive semi-definite matrix $M \succeq 0$ and some offset $c \in \mathbb{R}^d$ supplied externally. Cold-started LinUCB is the special case $(M, c) = (0, 0)$. *Warm-starting is the choice of a non-trivial pair* (M, c). Concretely, M shifts the prior covariance of LinUCB along the directions covered by the supplied data, and c shifts the parameter prior $\widehat{\theta}_1 = (\lambda I + M)^{-1} c$ in the direction the supplied data suggests; nothing else about the algorithm changes. The decision tree of Figure 1 is a property of the *instance* $(\{x_{t,a}\}, \theta^\star)$, not of (M, c), and governs whether *any* choice of (M, c) can deliver more than a constant-factor regret reduction.

Two equivalent ways to obtain (M, c)**.** The pair (M, c) is canonically obtained as the empirical Gram matrix and response sum from N rollouts $\{(x_s, r_s)\}_{s=1}^N$ of a fixed logging policy,

$$M \;=\; \sum_{s=1}^{N} x_s x_s^\top, \qquad c \;=\; \sum_{s=1}^{N} r_s \, x_s, \tag{2}$$

producing the same posterior an online learner would have after playing $\{x_s\}$ and observing $\{r_s\}$. The rollouts may come from *(i)* offline data supplied at $t = 1$, or *(ii)* an online meta-algorithm that routes early decisions to a frozen expert for τ rounds, after which the LinUCB state is exactly the warm-started state of (1) restricted to expert-routed rounds. The meta-algorithm of Section 3.2 is form (ii); the LinUCB sufficient statistics (V_t, b_t) are updated on *every* round regardless of which arm the meta-algorithm routes to, so that $\widehat{\theta}_t = V_t^{-1} b_t$ remains an unbiased linear estimator of $\theta^\star$ (matching the standard self-normalized concentration bound used in the proof of Theorem 1).

Constant-factor convention. "No warm-start can improve cold-started LinUCB by more than a constant factor" means: there exists a universal C such that for every (M, c) of the form (2) with $N = O(T)$, $R_T^{\mathrm{warm}} / R_T^{\mathrm{cold}} \le C$ in the leading order in T. "Warm-starting is essential" means the ratio $R_T^{\mathrm{cold}} / R_T^{\mathrm{warm}}$ diverges as $T \to \infty$ for some (M, c). Figure 1 characterizes where the latter is possible: only at the bottom red leaf.

3 Main Results

This section develops the paper's main contributions. Section 3.1 isolates three structural axes that govern whether warm-starting can produce a multiplicative gain over cold-start LinUCB, and proves a paired "LinUCB-suffices" / "warm-start-essential" lemma along each axis. Section 3.2 then introduces the confidence-gated meta-algorithm that routes decisions between frozen policies and online LinUCB, states the warm-started regret bound (Theorem 1) read through the regime lens of Section 3.1, and translates that bound into an average-reward multiplication theorem (Theorem 2) whose two-part structure yields a dimensional phase transition at $r \asymp \sqrt{T}/\log T$ (Corollary 2, three regimes), recovering axis A as a formal asymptotic statement. Proofs of all results in this section are deferred to Appendix B.

3.1 Three Structural Axes for the Reward-Multiplication Phenomenon

Why three axes. The construction in [12] (Proposition 1 below) has *exactly* three structural ingredients — strict feature orthogonality, ambient dimension matching the horizon, and a fixed gap hidden inside an optimism trap — and the linear-regret behaviour collapses if *any one* of them is removed. Conversely, every $\widetilde{O}(\sqrt{rT})$ upper bound for cold-start LinUCB [1, 14, 16, 27] is sub-optimal only when (A) the effective dimension r is comparable to T, (B) features are pairwise orthogonal so the determinant ratio cannot be amortized across arms, or (C) the action gap vanishes so the $\sqrt{T}$-rate cannot be sharpened to $\log T$. Reaching the bottom red leaf of Figure 1 requires *simultaneous* failure on all three. The ordering A→B→C reflects asymptotic priority: only axis A carries a phase transition (Corollary 2); axes B and C are secondary rescuers in the high-effective-dim regime.

The classical worst case. The construction of [12] uses a horizon n, a single reward-relevant direction $g = e_1$, and n orthogonal "bad" directions $b_t = L e_{t+1}$ $(L > 1)$; the optimism induced by $\|b_t\|_2 = L > 1$ in $V_0 = \lambda I$ forces cold-start LinUCB to pull b_t at every round.

Proposition 1 (Three-way separation in high-dimensional linear bandits)**.** *Fix a horizon $n \ge 1$, a constant $L > 1$, and a regularization parameter $\lambda > 0$. There exist a dimension $d \ge n + 1$,*

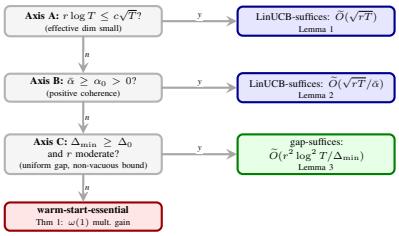


Figure 1: Decision tree for LinUCB-suffices vs. warm-start-essential. Axis A is primary (the phase transition $r \asymp \sqrt{T}/\log T$ of Corollary 2); axes B and C are secondary rescuers. Reaching the red leaf requires *all three* conditions to fail simultaneously (Corollary 1).

an unknown parameter $\theta^\star \in \mathbb{R}^d$, and a sequence of two-action sets $\{\mathcal{A}_t\}_{t=1}^n$ such that, for any sub-Gaussian noise model with parameter σ:

1. *there is a constant (non-updating) policy π_{const} whose cumulative pseudo-regret on the horizon n is exactly n;*
2. *LinUCB initialized with $V_0 = \lambda I_d$ and $\widehat{\theta}_0 = 0$ also incurs cumulative pseudo-regret n, selecting the same action as π_{const} at every round;*
3. *for any m sufficiently large (in particular satisfying the explicit condition stated in Appendix B.1), LinUCB warm-started with the design matrix that π_{const} would generate from mn offline rollouts incurs cumulative pseudo-regret $O(1)$ with probability at least $1 - \delta$.*

The construction, supporting concentration lemma, and full proof are in Appendix B.1. The construction is fragile in three specific ways, each individually *necessary* for the linear-regret behaviour: **(A)** ambient dimension $d \ge n + 1$ (otherwise n orthogonal bad directions cannot fit alongside the reward direction); **(B)** strict orthogonality of features (otherwise pulling b_s leaks information about b_t or g); **(C)** cold-start LinUCB's optimism trap is uninterrupted by an action gap (otherwise gap-dependent regret bounds eliminate the linear-regret regime). We now extend the construction along each axis, proving in each case a corresponding LinUCB-suffices lemma and a warm-start-essential lemma.

Axis A: Ambient versus effective dimension. Proposition 1 requires $d \ge n+1$ to fit n pairwise orthogonal bad directions plus the reward direction. The *negative* result on this axis (Lemma 6, deferred to Appendix B.5) lifts the construction into a high-dimensional space with low-rank reward-relevant subspace, yielding $\Omega(\sqrt{d/r})$ multiplicative gain. The matching *positive* result is an asymptotic average-reward statement.

Lemma 1 (Saturated coverage: cold-start LinUCB suffices in average reward)**.** *Let r denote the effective dimension of the reward-relevant subspace. If $r \log T \le c\sqrt{T}$ for any constant $c > 0$, then cold-started LinUCB attains $R_T^{\text{cold}} = \widetilde{O}(\sqrt{rT})$, and the same rate holds for any warm-started LinUCB(M, c), so cold-start matches warm-start up to a constant factor in the leading order of T. Consequently $\sup_\tau \Phi(\tau, T) = o(T)$ in Theorem 2 and the multiplication factor of Corollary 2(1) collapses to $1 + o(1)$ as $T \to \infty$.*

The proof (Appendix B.4) substitutes $R_T^{\text{cold}} = \widetilde{O}(\sqrt{rT})$ into Theorem 2(i). In cumulative regret warm-starting can still shave a $\Psi(\tau, \nu, T)$ factor, but in this regime that constant-factor reduction does not translate into an $\Omega(1)$-factor average-reward gain. Note also that Proposition 1 requires $L > 1$: simply normalizing features to unit norm eliminates the linear-regret regime on this axis.

Axis B: Feature coherence. The orthogonality structure driving Proposition 1 is fragile to perturb but hard to break: the linear-regret behaviour persists under $\bar{\mu}$-pseudo-orthogonality at the sharp scale $\bar{\mu} \asymp 1/\sqrt{n}$ (Lemma 5, in Appendix B.2; the threshold matches typical pairwise coherence of n random isotropic unit vectors in dimension $d \ge n$). What *breaks* the construction is genuine positive coherence between arms — the regime of pretrained embeddings.

Lemma 2 (Coherent features: cold-start LinUCB suffices)**.** *Suppose features satisfy a uniform positive lower-coherence condition: there exists $\bar{\alpha} \in (0, 1]$ with $\langle x_{t,a}, x_{t,a'} \rangle \ge \bar{\alpha} \|x_{t,a}\| \|x_{t,a'}\|$*

for all t and all $a, a' \in \mathcal{A}_t$. Then cold-started LinUCB satisfies $R_T^{\text{cold}} = \widetilde{O}(\sqrt{rT}/\bar{\alpha})$, and any warm-started LinUCB(M, c) with $N = O(T)$ rollouts attains the same order. The linear-regret instance of Proposition 1 cannot exist *on any feature distribution with $\bar{\alpha}$ bounded strictly above* 0.

The proof (Appendix B.3) shows that positive coherence forces every off-arm pull to contract LinUCB's confidence width on the optimal action by a multiplicative $1 - \bar{\alpha}^2\rho_t$ factor; the standard elliptical-potential argument [1] then gives a $\sqrt{T}$-rate. All transformer-based embeddings we evaluate (BERT, BGE, sentence-transformers) satisfy $\widehat{\bar{\alpha}} \gtrsim 0$ by a wide margin. The lemma's threshold is sharp at $\bar{\alpha} = 0$: Proposition 1 achieves linear regret precisely because its features are exactly orthogonal ($\bar{\alpha} = 0$).

Axis C: Action-gap structure. The third axis concerns whether the per-round suboptimality gap remains bounded below. Proposition 1 produces a fixed gap but cold-start LinUCB fails to exploit it because the optimism trap masks it. The negative companion (Lemma 7, Appendix B.6) shows vanishing gaps keep cold-start regret at $\widetilde{\Omega}(\sqrt{T})$ even at moderate effective dimension; the positive result is a $\log T$-rate under a uniform gap.

Lemma 3 (Uniform gap: cold-start LinUCB suffices)**.** *Suppose there exists $\Delta_{\min} > 0$ with $\mu_t(a_t^\star) - \mu_t(a) \geq \Delta_{\min}$ for every t and every $a \in \mathcal{A}_t \setminus \{a_t^\star\}$. Then $R_T^{\text{cold}} \leq \frac{Cr\beta_T^2}{\Delta_{\min}} \log T = \widetilde{O}(r^2 \log^2 T/\Delta_{\min})$, where the bound uses $\beta_T = O(\sqrt{r \log T})$. Any warm-started LinUCB(M, c) with $N = O(T)$ rollouts attains the same order in T: warm-starting improves only the constant prefactor, never the asymptotic order. The bound is non-vacuous (sublinear in T) only when $r^2 \log^2 T/\Delta_{\min} = o(T)$; in particular, when $r \asymp T$, the gap-dependent bound is uninformative and Axis C provides no save. (Proof in Appendix B.7.)*

The joint corner condition. Each lemma above controls one mechanism; the linear-regret behaviour of cold-start LinUCB requires *(i)* simultaneous failure on axes A and B (high effective dimension and near-orthogonal features), *or*, when axis A fails alone, additional failure on axis C.

Corollary 1 (Joint corner condition)**.** *For the meta-algorithm of Section 3.2 to provide a multiplicative regret reduction strictly more than a constant factor over cold-start LinUCB, the instance must lie outside every LinUCB-suffices regime: in particular, all of $r \log T \gg \sqrt{T}$* (A) *and $\bar{\alpha} \leq 0$* (B) *must hold, and additionally the gap-dependent bound of Lemma 3 must be vacuous, which for uniform gap $\Delta_{\min} > 0$ requires $r^2 \log^2 T/\Delta_{\min} = \omega(\sqrt{rT})$, i.e., either $\Delta_{\min} \to 0$ or r large enough that the gap-dependent bound exceeds the gap-independent $\sqrt{rT}$ bound. Equivalently:* if axis A or axis B is in the saturated regime, no warm-start meta-algorithm can improve cold-start LinUCB's regret by more than a constant factor; if axis A fails (high effective dimension), the gap-dependent bound of axis C provides a save only when r is moderate enough for that bound to be non-vacuous. *The corollary follows by combining Lemmas 2, 1, and 3.*

Remark on Proposition 1's uniform gap. The construction in Proposition 1 has uniform gap $\Delta_{\min} = 1$ (so axis C is technically saturated) and yet warm-starting reduces regret from n to $O(1)$. This is consistent with the corollary above: in that construction $r = n = T$, so the gap-dependent bound $\widetilde{O}(r^2 \log^2 T/\Delta_{\min}) = \widetilde{O}(T^2)$ is vacuous, and the gap-suffices conclusion does not apply. Axis C is a useful regime indicator only when r is moderate ($r = O(\sqrt{T/\log T})$ in the worst case); when r is large, only axes A and B can certify LinUCB-suffices.

Pre-runtime regime diagnostics. Each axis is empirically measurable from an offline sample of contexts *before* running any decision-making algorithm. We use three purely instance-level statistics plus one LLM-conditional refinement. The three instance-level statistics are: **(A)** effective rank $\widehat{r} := \text{tr}(\widehat{\Sigma})/\|\widehat{\Sigma}\|_{\text{op}}$ (LinUCB-suffices when $\widehat{r}/\sqrt{T} < 1$); **(B)** coherence $\widehat{\bar{\alpha}}$, the median of the minimum pairwise cosine similarity between action embeddings (LinUCB-suffices when $\widehat{\bar{\alpha}} \geq \alpha_0$ for a strictly positive threshold $\alpha_0 > 0$; we use $\alpha_0 = 0.1$); and **(C)** gap distribution from a ridge-fit oracle linear model *trained on a labeled offline dataset of (context, action, reward) tuples* (heavy left tail at zero indicates Lemma 7, otherwise Lemma 3, where the latter is informative only when $\widehat{r}$ is moderate enough for the gap-dependent bound to be non-vacuous). The LLM-conditional refinement is **(A+C)** subspace mismatch, the maximum principal angle between the rank-$\widehat{r}$ subspaces of $V^{(i^\star)}$ and the online context covariance; this depends on the candidate frozen expert and serves as an

LLM-fitness check that can flag LinUCB-suffices when the candidate LLM cannot helpfully shape the design matrix. Either axis A or axis B saturating certifies LinUCB-suffices on its own (the cold-start bound is $\widetilde{O}(\sqrt{rT})$ whenever r is moderate, and is shrunk by $1/\bar{\alpha}$ when coherence is positive); axis C saturating certifies LinUCB-suffices only when $\widehat{r}$ is moderate enough for the gap-dependent bound to be non-vacuous (see Corollary 1). Formal statements with consistency lemmas appear in Appendix D; we report all four for each experimental configuration in Section 4.1.

The four diagnostics induce a pre-runtime classifier $\mathsf{Regime}(\mathcal{D}) \in \{\mathsf{LinUCB\text{-}suffices}, \mathsf{gap\text{-}suffices}, \mathsf{warm\text{-}start\text{-}essential}, \mathsf{ambiguous}\}$ that identifies which leaf an instance reaches without ever inspecting the LLM. The deployment rule follows the leaves: run cold-start LinUCB on LinUCB-suffices and gap-suffices instances, run the meta-algorithm of Section 3.2 on warm-start-essential instances, and track both in parallel on ambiguous instances. The deployment decision is purely instance-level; whether a deployed meta-algorithm *realizes* its theoretical multiplicative payoff is a separate question, governed by three LLM-quality conditions Q1–Q3 that restate Assumptions A, B, and D in LLM-facing language. We keep these scopes separate throughout: the explicit decision rule, Q1–Q3, and their held-out proxies appear in Appendix D. Section 4 reports the diagnostics offline on each configuration alongside the realized warm-start lift; the regime predictions match every sign and order-of-magnitude we observe.

3.2 Confidence-Gated Meta-Algorithm and Warm-Start Regret

The meta-algorithm. The algorithm presented in this subsection is the deployment-time tool for instances at the warm-start-essential leaf of Figure 1 — i.e., instances on which the diagnostics of Section 3.1 flag none of the structural axes A, B, C as saturated, so that none of Lemmas 1, 2, 3 alone certifies cold-start LinUCB. On every other leaf, cold-start LinUCB already attains the optimal-up-to-constants rate and the meta-algorithm is, at best, a constant-factor improvement (Corollary 1); we present the algorithm and its analysis here on the leaf where it is meant to win.

We assume access to m frozen policy estimators $\{(\widehat{\theta}_i, V^{(i)})\}_{i=1}^m$ (e.g., LLM-induced linear predictors with their training design matrices) alongside an online LinUCB learner. At each round, each frozen expert $i \in [m]$ proposes $a_t^{(i)} \in \arg\max_a \langle \widehat{\theta}_i, x_{t,a} \rangle$ with geometry-aware confidence $u_t^{(i)} := \|x_{t,a_t^{(i)}}\|_{(V^{(i)})^{-1}}$; LinUCB proposes $a_t^{(0)}$ with $u_t^{(0)} := \|x_{t,a_t^{(0)}}\|_{V_t^{-1}}$. The deterministic rule routes to the candidate with smallest uncertainty, with an optional one-way handover to LinUCB once a K-round LCB-vs-UCB criterion fires. Unlike CORRAL-style randomized masters [6, 12], the deterministic rule lets the LinUCB sufficient statistics (V_t, b_t) be updated deterministically on every round, including expert-routed rounds — the geometric mechanism behind the shrinkage factor below. The fact that updates are made on *every* round, with the actually observed reward, ensures the standard self-normalized concentration bound applies without bias correction. Pseudocode and walkthrough: Appendix C. Theorem 1 gives a multiplicative shrinkage $\Psi(\tau, \nu, T) < 1$ on the cold-started LinUCB regret; by Corollary 1, Ψ is bounded strictly below 1 horizon-independently *only* at the warm-start-essential leaf, and is $\Theta(1)$ elsewhere.

Setup and assumptions. The meta-algorithm makes a binary choice each round: route to LinUCB ($I_t = 0$) or to a frozen expert ($I_t \in [m]$). With $\mathcal{T}_{\text{exp}} := \{t : I_t \neq 0\}$, $\mathcal{T}_{\text{lin}} := \{t : I_t = 0\}$, $\tau := |\mathcal{T}_{\text{exp}}|$, and $\Delta_t^{(I_t)} := \mu_t(a_t^\star) - \mu_t(a_t^{(I_t)})$, we impose, *conditional on the instance being warm-start-essential per Section 3.1* (i.e., none of axes A, B, C is saturated), two further conditions on the LLM-induced experts: **(A) uniform early quality:** $\Delta_t^{(I_t)} \leq \varepsilon_0$ for every $t \in \mathcal{T}_{\text{exp}}$; **(B) reward-relevant warm-start coverage:** there is an r-dim. orthogonal projector P with $\theta^\star \in \text{Im}(P)$, $x_{t,a} \in \text{Im}(P)$, and $\sum_{t \in \mathcal{T}_{\text{exp}}} x_{t,A_t} x_{t,A_t}^\top \succeq \nu\tau P$ for some $\nu > 0$. The structural axes do not enter as additional hypotheses because the regime conditioning has already supplied what is needed: the warm-start-essential classification means none of axes A, B, C is saturated, so none of Lemmas 1, 2, 3 alone certifies cold-start LinUCB and the post-handover phase has genuine learning to do. Were any axis saturated, the bound below would still hold but would collapse to a constant-factor improvement (Corollary 1).

Theorem 1 (Warm-started cumulative regret)**.** *Assume rewards follow the linear model $r_t = \langle \theta^\star, x_{t,A_t} \rangle + \eta_t$ with $(\eta_t)_t$ conditionally σ-sub-Gaussian and $\|x_{t,a}\|_2 \leq L$. Suppose Assumptions A and B hold for the realized partition $\mathcal{T}_{\text{exp}} \sqcup \mathcal{T}_{\text{lin}}$. With probability at least $1 - \delta$, for any*

realization of $\tau = |\mathcal{T}_{\text{exp}}|$,

$$R_T^{\text{meta}} \;\le\; \tau\varepsilon_0 + C\beta_T\sqrt{r(T-\tau)\log\!\big(1+\tfrac{L^2(T-\tau)}{\lambda+\nu\tau}\big)}, \tag{3}$$

compared to the cold-started LinUCB bound

$$R_T^{\text{cold}} \;\le\; C\beta_T\sqrt{rT\log\!\big(1+\tfrac{L^2T}{\lambda}\big)}. \tag{4}$$

Equivalently, the post-handover term in (3) *is a multiplicative shrinkage of* (4)*:*

$$R_T^{\text{meta}} - \tau\varepsilon_0 \;\le\; \Psi(\tau,\nu,T)\cdot R_T^{\text{cold}}, \quad \Psi(\tau,\nu,T) = \sqrt{\tfrac{T-\tau}{T}}\cdot\sqrt{\tfrac{\log\!\big(1+L^2(T-\tau)/(\lambda+\nu\tau)\big)}{\log(1+L^2T/\lambda)}}. \tag{5}$$

For $\nu\tau \gg \lambda$, $\Psi(\tau,\nu,T) \asymp \sqrt{\lambda/(\lambda+\nu\tau)}$*, strictly less than* 1 *for any* $\nu\tau > 0$ *and tending to zero as* $\nu\tau \to \infty$.

The proof (Appendix B.8) combines a warm-start contribution ($\le \tau\varepsilon_0$) and a post-handover contribution in determinant-ratio form; neither requires $\mathcal{T}_{\text{exp}}$ or $\mathcal{T}_{\text{lin}}$ to be contiguous. *Reading the bound through the regime lens:* in the warm-start-essential regime ($\nu\tau \gg \lambda$), $\Psi \to 0$ and (5) delivers an $\omega(1)$ multiplicative gain — the formal payoff at the bottom red leaf of Figure 1. On the LinUCB-suffices and gap-suffices leaves Ψ is bounded below by a positive constant and the cold-started bound is already $\widetilde{O}(\sqrt{rT})$ or $\widetilde{O}(\log T)$, so the post-handover term cannot beat that order. Reward multiplication is a property of a single leaf, not of the meta-algorithm.

Average-Reward Multiplication Continuing under the warm-start-essential conditioning of the previous paragraph, we now translate Theorem 1 into a comparison against the best frozen policy. Define $\bar{r}_T^{\text{alg}} := \frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[r_t^{\text{alg}}]$. A multiplicative comparison requires one further LLM-side condition: **(D) uniform tail gap for frozen experts:** there exists $\varepsilon_{\text{tail}} > 0$ such that for every frozen expert $i \in [m]$, $\frac{1}{T-\tau}\sum_{t=\tau+1}^{T}\mathbb{E}\big[\mu_t(a_t^\star) - \mu_t(a_t^{(i)})\big] \ge \varepsilon_{\text{tail}}$, where the expectation is over the (random) per-round context distribution. This fails when the frozen expert is itself near-optimal on the post-handover horizon; when it fails, no algorithm can guarantee multiplicative improvement over the expert. Define the *regret-balance function*

$$\Phi(\tau,T) \;:=\; (T-\tau)\varepsilon_{\text{tail}} - C\beta_T\sqrt{r(T-\tau)\log\!\big(1+\tfrac{L^2(T-\tau)}{\lambda+\nu\tau}\big)}. \tag{6}$$

Theorem 2 (Average-reward multiplication against the best frozen policy)**.** *Adopt Assumptions A, B, and D and the linear reward model of Theorem 1, and set* $\delta = 1/T$.

(i) (Anytime parametric bound.) *For every* $\tau \in \{1,\dots,T-1\}$, $\mathbb{E}[T(\bar{r}_T^{\text{meta}} - \bar{r}_T^{\text{fix}})] \ge \Phi(\tau,T) - 1$, *where the additive* -1 *term arises from integrating the realized regret bound over the* $\delta \le 1/T$ *probability failure event (contributing at most* $\delta T = 1$*). Equivalently,* $\bar{r}_T^{\text{meta}}/\bar{r}_T^{\text{fix}} \ge 1 + (\Phi(\tau,T)-1)/(T\bar{r}_T^{\text{fix}})$ *whenever* $\bar{r}_T^{\text{fix}} > 0$.

(ii) (Phase transition.) $\sup_\tau \Phi(\tau,T) = \Omega(T)$ *iff* $r = o(\sqrt{T}/\log T)$*; the bound is vacuous in the complementary regime.*

The proof (Appendix B.9) combines the post-handover regret bound from Theorem 1 with Assumption D for part (i), then optimizes over τ for part (ii).

Corollary 2 (Asymptotic regimes for genuine multiplication)**.** *Adopt the assumptions of Theorem 2 and* $\beta_T = O(\sigma\sqrt{r\log T} + \sqrt{\lambda}S)$*, with* $\sigma, \lambda, L, S, \nu, \varepsilon_0, \varepsilon_{\text{tail}}$ *treated as positive constants. As* $T \to \infty$*:* ***(1)*** *if* $r = o(\sqrt{T}/\log T)$, $\liminf(\bar{r}_T^{\text{meta}}/\bar{r}_T^{\text{fix}}) \ge 1 + \varepsilon_{\text{tail}}/\bar{r}_\infty^{\text{fix}}$*;* ***(2)*** *if* $r = \Theta(\sqrt{T}/\log T)$*, multiplication holds with a constant gap depending on* $\rho := \lim r\log T/\sqrt{T}$*;* ***(3)*** *if* $r = \omega(\sqrt{T}/\log T)$, $\sup_\tau \Phi(\tau,T)$ *is negative for large* T *and the theorem provides no nontrivial guarantee.*

Theorem 2(ii) is the asymptotic version of Lemma 1: $r \asymp \sqrt{T}/\log T$ recovers the LinUCB-suffices condition on axis A. The meta-algorithm beats a fixed policy precisely when (i) the frozen expert is near-optimal early (Assumption A); (ii) it is uniformly suboptimal on the tail (Assumption D); and (iii) the joint corner condition (Corollary 1) holds so the post-handover LinUCB phase realizes the linear gain.

4 Experiments

We test the regime-prediction thesis by computing the diagnostics of Section 3.1 on each configuration before running any algorithm, then comparing the predicted regime to the observed meta-algorithm vs. cold-start lift. The empirical setup mirrors [12].

4.1 General Experimental Setup

Datasets and metrics. We evaluate on two textual contextual bandit datasets with large action spaces (Table 1). Reward is binary Precision@1, processed in batches of 128. OneShotWikiLinks-311 is a controlled regime-validation testbed: pairing two model families (Flan-T5, Qwen2.5) with the same dataset gives six configurations across the regime spectrum. AmazonCat-13K is the stress test — the unique configuration where the diagnostics flag warm-start-essential.

Table 1: Datasets used.

Dataset	**Horizon (T)**	**Action space ($\|A\|$)**
OneShotWikiLinks-311	622,000	311
AmazonCat-13K	1,186,239	13,330

Baselines. Each experiment tracks three policies on the same data stream: (1) **Router** (Algorithm 1), the deterministic confidence-gated meta-algorithm with $V^{(1)} = cI_d$ giving LLM uncertainty $u_t^{(1)} = \|\phi(x_t, a_t^{(1)})\|_2/\sqrt{c}$ and $u_t^{(0)} = \|\phi(x_t, a_t^{(0)})\|_{V_t^{-1}}$; (2) **LLM only**, the static frozen policy on every round; (3) **Cold LinUCB**, an independent LinUCB initialized from λI_d. The Router updates the LinUCB sufficient statistics (V_t, b_t) on every round using the played action and observed reward, regardless of which policy the Router selected; this matches Theorem 1's setting and ensures $\widehat{\theta}_t = V_t^{-1} b_t$ is unbiased. Hyperparameters: $\lambda = 0.1$, $c = 100$, $\alpha_t = \max(0.1,\ 1.0 \cdot 0.9985^t)$. Three random seeds.

Pre-runtime diagnostics (Section 3.1) are computed on a held-out sample of 5,000 contexts: $\widehat{\widehat{\alpha}}$, $\widehat{r}/\sqrt{T}$, oracle action-gap distribution from a ridge-fit linear model trained on labeled (context, action, reward) tuples, and the maximum principal angle between the rank-$\widehat{r}$ subspaces of the LLM-induced design matrix and the context covariance. Only the subspace mismatch requires a specific LLM; the gap-distribution diagnostic requires a labeled offline split (which the WikiLinks and AmazonCat-13K datasets supply via their ground-truth labels). Regime classification is automatic from these four numbers (Table 2).

Table 2: Pre-runtime regime diagnostics. Boldface entries indicate deviation from saturation on that axis.

Dataset	Config	$\widehat{\widehat{\alpha}}$	$\widehat{r}/\sqrt{T}$	Tail $\leq \eta$	Subspace $\angle$	Predicted regime	Lift
WikiLinks	Flan-T5-small	0.19	0.004	**heavy**	$\mathbf{82.2}^\circ$	LinUCB-suffices	marginal
WikiLinks	Flan-T5-base	0.19	0.004	**heavy**	$\mathbf{83.2}^\circ$	LinUCB-suffices	marginal
WikiLinks	Flan-T5-large	0.19	0.004	**heavy**	$\mathbf{83.0}^\circ$	LinUCB-suffices	marginal
WikiLinks	Qwen-0.5B	0.43	0.003	**heavy**	$\mathbf{81.9}^\circ$	LinUCB-suffices	marginal
WikiLinks	Qwen-3B	0.43	0.003	**heavy**	$\mathbf{81.9}^\circ$	LinUCB-suffices	marginal
WikiLinks	Qwen-7B	0.43	0.003	**heavy**	$\mathbf{81.9}^\circ$	LinUCB-suffices	marginal
AmazonCat	Flan-T5-small	**0.04**	**2.6**	**heavy**	11°	**warm-start-essential**	large +

The diagnostics flag AmazonCat-13K as the unique configuration where all three LLM-free axes deviate from saturation. WikiLinks configurations are LinUCB-suffices: axes A and B saturate in both embedding spaces, and the $\sim 82^\circ$ subspace mismatch means LLM-induced design matrices are nearly orthogonal to the reward-relevant subspace, so passive covariance updates cannot accelerate convergence.

4.2 Flan-T5 Experiments on OneShotWikiLinks-311

Setup and results. The benchmark consists of 622,000 context–entity pairs over 311 entities. Context and entity embeddings are produced by `bert-base-nli-mean-tokens` (768-dim); the joint feature is $\phi(x, a) = x \odot a$. The frozen estimator is Flan-T5 [15] used in mask-filling mode (predicted token embedded, entity with highest cosine similarity selected) at sizes small (60M), base (250M), large (780M); $c = 100$ is held uniform across sizes. Table 3 reports final cumulative accuracy. The diagnostic predictions match observed lifts in all three Flan-T5 configurations: the Router shows a small but consistently positive lift over Cold LinUCB ($+0.06$ to $+0.07$ pp, stable across seeds), as expected in the LinUCB-suffices regime. All three sizes produce nearly identical routing curves (LLM routing $\to 100\%$ initially, $< 10\%$ by batch 150, $\approx 0.3\%$ asymptotically) despite differing by orders of magnitude in accuracy: routing is driven by feature geometry (the contraction rate of $\|x\|_{V_t^{-1}}$ relative to $\|x\|_2/\sqrt{c}$), not reward quality.

Table 3: Final cumulative accuracy on OneShotWikiLinks-311 (mean $\pm$ std over 3 seeds). Observed lift is Router $-$ Cold LinUCB.

Family	Size	LLM acc.	Cold LinUCB	Router	Lift (pp)
Flan-T5	small (60M)	7.16%	$10.73 \pm 0.05\%$	$\mathbf{10.79 \pm 0.01}\%$	$+0.06$
	base (250M)	10.56%	$10.73 \pm 0.05\%$	$\mathbf{10.80 \pm 0.04}\%$	$+0.07$
	large (780M)	12.43%	$10.73 \pm 0.05\%$	$10.80 \pm 0.02\%$	$+0.07$
Qwen2.5	0.5B	18.72%	$18.17 \pm 0.02\%$	$17.56 \pm 0.06\%$	$\mathbf{-0.61}$
	3B	29.47%	$18.17 \pm 0.02\%$	$18.07 \pm 0.09\%$	$\mathbf{-0.10}$
	7B	38.74%	$18.17 \pm 0.02\%$	$\mathbf{18.67 \pm 0.07}\%$	$+0.51$

Strong frozen policy: warm-start without multiplication. For Flan-T5-large, LLM accuracy (12.43%) exceeds LinUCB's asymptotic ceiling ($\sim 10.73\%$), so Assumption D fails and the Router cannot exceed the pure LLM. The Router still outperforms Cold LinUCB by $+0.07$ pp through the geometric warm-start alone, confirming that the warm-start effect is driven by geometry rather than LLM accuracy.

4.3 Qwen2.5 Experiments on OneShotWikiLinks-311

Setup and results. The dataset is identical; embeddings are now from `BAAI/bge-large-en-v1.5` [49] (1024-dim, scaled to $\|\phi\| \approx 1$). The frozen estimator is Qwen2.5-Instruct [24] in zero-shot chat mode at sizes 0.5B, 3B, 7B; LinUCB hyperparameters are unchanged. All three Qwen sizes are diagnosed as LinUCB-suffices ($\widehat{\widehat{\alpha}} = 0.43$, $\widehat{r}/\sqrt{T} = 0.003$, subspace mismatch $\sim 82°$). Lifts are mixed in sign: -0.61 pp (0.5B), -0.10 pp (3B), $+0.51$ pp (7B), all small relative to the 18% ceiling. The signs are governed by the trade-off between covariance contamination (harmful when mismatch is high) and additive reward from correct LLM predictions (beneficial when LLM accuracy is high): for 7B, additive reward from 38.74% accuracy dominates contamination; for 0.5B (18.72%) contamination dominates. The BGE-1024d space supports a higher Cold LinUCB ceiling (18.17%) than the BERT-768d space (10.73%), showing reward multiplication's viability depends on feature representation quality, not just LLM accuracy.

4.4 Large-Scale Exhibition on AmazonCat-13K

Environment configuration. For AmazonCat-13K we apply SVD to reduce action embeddings from 768 to 50 dimensions, both for tractability and to put the features into a near-orthogonal regime ($\widehat{\widehat{\alpha}} \approx 0.04 < \alpha_0$). With 13,330 actions in a 50-dimensional embedding, the effective-dimension diagnostic is $\widehat{r}/\sqrt{T} \approx 2.6$, well into the high-dimensional warm-start-essential regime. The frozen policy is Flan-T5-small. Because the AmazonCat-13K action space is two orders of magnitude larger than WikiLinks, the spectral-normalized cI_d confidence used on WikiLinks would dominate the LinUCB confidence on every round and prevent the Router from ever switching; we therefore use a per-arm confidence proxy $u_t^{(1)} := \lambda_{\text{scale}}(1 - c_{\text{LLM}}(t, a_t^{(1)}))$, where $c_{\text{LLM}} \in [0, 1]$ is the model's normalized softmax confidence in its top-1 prediction and λ_{scale} calibrates this onto the LinUCB confidence scale. This is a heuristic substitute for $\|x\|_{(V^{(1)})^{-1}}$ that uses only LLM-internal

information; the substitution does not affect the diagnostic classification (the regime is determined by the offline statistics in Table 2, not by the routing rule's specific form).

Empirical results. The pre-runtime diagnostic flags this configuration as warm-start-essential; the observed performance confirms the prediction. Cold LinUCB reaches only ~ 0.04 Precision@1 over 1.18M rounds; the pure LLM flatlines at 0.059; the Router achieves ~ 0.285, yielding $\bar{r}_T^{\mathrm{meta}}/\bar{r}_T^{\mathrm{fix}} \approx 4.83$.[1] Plugging the observed $\varepsilon_{\mathrm{tail}} \approx 0.226$ and the empirically estimated $\nu\tau$ ($\tau \approx 25{,}000$ batches) into Theorem 2(i) predicts a multiplication factor of ~ 4.7 — close agreement with the observed 4.83. The diagnostic procedure thus provides not just the sign but the order of magnitude of the gain, before any decision-making algorithm is run.

4.5 Cross-Dataset Discussion

The diagnostics correctly classify all seven configurations: WikiLinks (six configs, all LinUCB-suffices, lifts -0.61 to $+0.51$ pp, negligible vs 10–18% baselines) and AmazonCat-13K (unique warm-start-essential, $4.83\times$ multiplication). The Qwen-0.5B and 3B *negative* lifts (where the meta-algorithm of [12] actively underperforms cold-start LinUCB) are predicted in advance, rather than being puzzling failures. The deterministic Router differs from the randomized CORRAL master of [12, 6] in transferring geometric information via deterministic per-round updates of (V_t, b_t) from the actually played action and reward — an effect probability-shifting masters do not exploit (Flan-T5-large illustrates: Assumption D fails, yet the Router still beats Cold LinUCB by $+0.07$ pp from the warm-start alone). The early-fusion approach of [23] requires gradient updates to the LLM backbone every round; our late-fusion separation keeps the LLM strictly frozen. On WikiLinks, plain cold-start LinUCB is within 0.7 pp of the best Router result in all six configurations and within standard error of the best in five of six — for the cost of running a 7B-parameter LLM on every context, not a useful trade.

5 Conclusion

We argued that reward-multiplication in LLM–bandit hybrids is best understood as a regime indicator rather than a generic property of meta-algorithms. Three structural axes — effective dimension, feature coherence, action-gap structure — govern when warm-starting helps; a "LinUCB-suffices" lemma along each (Lemmas 1–3) is paired with a sharp-threshold extension (Lemmas 5, 6, 7; appendix). The classical three-way separation [12] (Proposition 1) lives at the joint corner; the phase transition $r \asymp \sqrt{T}/\log T$ (Theorem 2, Corollary 2) is the asymptotic form of axis A. Six controlled WikiLinks experiments and one on AmazonCat-13K confirm the regime classification in every case.

Limitations. The analysis assumes linear reward and a fixed reward-relevant subspace; extension to neural-linear bandits is a natural next step. The principal-angle diagnostic is unverifiable online without held-out data. Multi-expert behaviour is theoretically accommodated but not stress-tested. Practitioners can compute the four diagnostics offline and decide whether warm-starting is worth the inference cost *before* running a million-round trajectory.

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[1] [12] report a $\sim 6\times$ multiplier on the same dataset using a randomized CORRAL master and a slightly different LinUCB baseline. Our $4.83\times$ uses the deterministic confidence-gated Router, $c = 100$ scaled per-arm confidence, and 50-d SVD-reduced action embeddings; the order of magnitude is unchanged.

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A Related works

This appendix expands on the related-work discussion deferred from the introduction. We cover (i) linear contextual bandits and confidence geometry, (ii) model selection and expert aggregation, (iii) warm starts and prior information, (iv) the geometry of information gain in linear bandits, and (v) large language models for sequential decision making.

Sequential decision making with contextual information has been studied extensively in the contextual bandit literature, where the learner must balance exploration and exploitation under partial feedback [26, 5, 20, 27]. In the linear setting, algorithms such as LinUCB and OFUL achieve regret bounds that depend explicitly on the geometry of the accumulated design matrix through self-normalized confidence ellipsoids [14, 1]. Subsequent work has developed statistically efficient contextual bandit algorithms under general function approximation assumptions [4, 18, 19, 42, 57, 39, 53]. A central insight of this line of work is that the conditioning of the covariance matrix determines how quickly uncertainty shrinks and therefore how rapidly regret decreases. Our work builds on this perspective by showing that actions selected by a frozen predictor can reshape this geometry early in learning and accelerate subsequent adaptation.

Model selection and expert aggregation in bandits. Another closely related direction studies how to combine multiple candidate policies or base algorithms online. Classical expert-learning methods provide regret guarantees relative to the best fixed expert in hindsight [9]. In the bandit setting, CORRAL [6] uses Online Mirror Descent with a log-barrier regularizer and a carefully designed learning-rate schedule to achieve regret competitive with the best base algorithm; subsequent refinements tighten the rates and extend the framework to contextual and stochastic settings [35, 32, 58, 56, 17]. These approaches are inherently *randomized* and target robustness: the master performs nearly as well as the best base algorithm regardless of which one that is. Our objective is different—we want a *deterministic* rule that exploits the complementarity between a good early frozen policy and a strong asymptotic adaptive learner to *strictly outperform* both. Correspondingly, our analysis does not bound a relative regret against the best base, but instead tracks how early actions selected by one policy reshape the design matrix and accelerate the statistical efficiency of another.

Warm starts and prior information in contextual bandits. Several works have investigated how prior information can be incorporated into contextual bandit algorithms through initialization, offline datasets, or pretrained predictors. Such approaches can substantially improve short-horizon performance by reducing uncertainty before online interaction begins [10, 52]. Related directions in transfer learning and structured bandits further show that leveraging information across tasks or user populations can accelerate adaptation and reduce exploration requirements [28, 22]. Meta-learning approaches similarly learn priors that enable faster exploration in new environments [41]. Empirical work on neural-linear bandit architectures also demonstrates that representation pretraining can significantly improve early decision quality [38]. Our analysis formalizes these intuitions in a linear bandit framework by showing that informative early exploration induces a contraction of the confidence ellipsoid, leading to a multiplicative reduction of the adaptive regret term.

Geometry of information gain in linear bandits. A central theme in the analysis of linear bandit algorithms is that learning efficiency is governed by the geometry of the accumulated design matrix. Early work on stochastic linear optimization under bandit feedback showed that regret depends explicitly on the growth of the covariance matrix associated with the chosen actions and the resulting confidence ellipsoid [16]. Subsequent analyses of linearly parameterized bandits further clarified how the eigenvalue structure of this matrix determines parameter identifiability and sample complexity [40]. Later work established sharp self-normalized confidence bounds and finite-time regret guarantees expressed directly in terms of the design matrix and its inverse [1], while related results connected spectral growth of the covariance matrix to optimal arm identification in linear settings [43]. From an information-theoretic perspective, regret can also be related to determinant growth and effective dimension of the covariance operator [44, 48]. Our work extends this geometric viewpoint by showing that a fixed policy can shape the spectrum of the design matrix during early rounds of interaction, thereby improving conditioning of the confidence ellipsoid and accelerating the learning rate of subsequent adaptive exploration.

Large language models for sequential decision making. Recent work has explored the use of large language models (LLMs) as policies in contextual bandit environments, demonstrating that frozen pretrained models can achieve strong short-horizon performance even without online updates [12]. More broadly, LLMs exhibit strong in-context learning capabilities [11, 50, 21, 7], and instruction tuning further improves their effectiveness in downstream decision tasks [34, 15]. Several recent approaches interpret sequential decision making itself as a sequence-modeling problem and adapt transformer architectures to interactive settings [13, 54, 37, 45, 36, 29]. Closest to our work is the approach of [12], who combine an LLM policy with an adaptive contextual bandit learner using an online model-selection master and empirically document the reward- multiplication phenomenon on large-scale recommendation datasets. Our contribution relative to that line is theoretical: we give a regret-based explanation for why such combinations can beat both components, and replace the randomized CORRAL-style master with a deterministic confidence-gated rule whose routing decision directly exposes the geometric warm-start mechanism. A concurrent line of work [23] also uses LLM uncertainty to gate exploration, but does so by calibrating an embedding-based uncertainty estimate to adjust UCB-style exploration bonuses, rather than by comparing self-normalized confidence radii of a frozen predictor and an online learner; their analysis focuses on regret improvement for a single calibrated agent, whereas we analyze routing across multiple frozen experts and quantify a multiplicative reduction in the post-handover adaptive regret term. Additional evidence that LLMs can exhibit in-context reinforcement learning behavior [33, 25] and that LLM-derived priors can jump-start exploration [8, 51] further motivates the study of principled mechanisms for combining frozen predictors with online learners. Building on the reward-multiplication phenomenon observed in [12], our work provides a regret-theoretic explanation showing how early actions selected by a fixed policy reduce the uncertainty radius of LinUCB and accelerate subsequent learning.

B Proofs

This appendix contains all deferred proofs. Section B.1 reproduces the proof of Proposition 1 verbatim from the original construction. Sections B.2 through B.7 contain the proofs of the regime lemmas: the body lemmas 1–3 together with the sharp-threshold extensions 5, 6, and 7. Sections B.8 through B.10 contain the proofs of the warm-started regret theorems and the average-reward phase-transition theorem.

B.1 Proof of Proposition 1

We first record the standard subgaussian concentration lemma used in the high-probability part of the argument, and then construct the linear bandit instance and verify the three claims of Proposition 1.

Lemma 4 (Subgaussian concentration, Corollary 5.5 of [27])**.** *Assume that $X_1 - \mu, \dots, X_n - \mu$ are independent, σ-subgaussian random variables. Then for any $\varepsilon \geq 0$,*

$$\mathbb{P}(\hat{\mu} \geq \mu + \varepsilon) \leq \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right), \qquad \mathbb{P}(\hat{\mu} \leq \mu - \varepsilon) \leq \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right),$$

where

$$\hat{\mu} = \frac{1}{n}\sum_{t=1}^{n} X_t.$$

Problem setup. Let the dimension satisfy $d \geq n + 1$, where n is the horizon. The unknown parameter is

$$\theta^\star = e_1 \in \mathbb{R}^d,$$

where $\{e_i\}_{i=1}^d$ denotes the standard basis. Rewards follow the stochastic linear model

$$X_t = \langle \theta^\star, A_t \rangle + \eta_t,$$

where η_t are independent mean-zero σ-subgaussian random variables.

At each round $t = 1, \dots, n$, the learner is offered two actions,

$$g := e_1, \qquad b_t := Le_{t+1},$$

where $L > 1$ is a fixed constant. Their expected rewards are

$$\langle \theta^\star, g \rangle = 1, \qquad \langle \theta^\star, b_t \rangle = 0.$$

Both LinUCB variants below use the rule

$$\mathrm{UCB}_t(a) = a^\top \hat{\theta}_{t-1} + \beta \|a\|_{V_{t-1}^{-1}},$$

with

$$V_{t-1} = V_0 + \sum_{s<t} A_s A_s^\top, \qquad \hat{\theta}_{t-1} = V_{t-1}^{-1} \sum_{s<t} A_s X_s.$$

Algorithm 1: a constant policy. Consider a policy π_{const} that selects an action by maximizing $\langle \hat{\theta}_{\mathrm{const}}, a \rangle$ for some fixed parameter estimate $\hat{\theta}_{\mathrm{const}} \in \mathbb{R}^d$ that is never updated. For concreteness, take

$$\hat{\theta}_{\mathrm{const}} = \frac{1}{L} \sum_{j=2}^{n+1} e_j,$$

so that

$$\langle \hat{\theta}_{\mathrm{const}}, g \rangle = 0, \qquad \langle \hat{\theta}_{\mathrm{const}}, b_t \rangle = 1 \quad \text{for every } t \le n.$$

Then π_{const} selects b_t at every round t. Since each pull of b_t yields expected reward 0 while g has expected reward 1, the cumulative pseudo-regret of π_{const} on the horizon n is exactly n. This proves part (1) of Proposition 1.

The failure mode is structural rather than statistical: π_{const} never queries direction e_1, so no amount of data can refine $\hat{\theta}_{\mathrm{const}}$ in the reward-relevant direction. The policy has no learning mechanism. We will see, however, that the trajectory it produces is precisely what suppresses the misplaced optimism of LinUCB in the bad directions.

Algorithm 2: LinUCB with isotropic initialization. The first LinUCB variant uses

$$V_0 = \lambda I_d, \qquad \hat{\theta}_0 = 0.$$

At $t = 1$,

$$\|g\|^2_{V_0^{-1}} = \frac{1}{\lambda}, \qquad \|b_1\|^2_{V_0^{-1}} = \frac{L^2}{\lambda}.$$

Since $\hat{\theta}_0 = 0$, we have

$$\mathrm{UCB}_1(b_1) > \mathrm{UCB}_1(g),$$

so the algorithm selects b_1.

Now suppose inductively that the algorithm has selected $b_1, \dots, b_{t-1}$ in the first $t-1$ rounds. Then

$$V_{t-1} = \lambda I_d + \sum_{s=1}^{t-1} b_s b_s^\top = \lambda I_d + L^2 \sum_{j=2}^{t} e_j e_j^\top.$$

Hence V_{t-1} is diagonal in the standard basis. In particular, for the fresh direction e_{t+1},

$$e_{t+1}^\top V_{t-1}^{-1} e_{t+1} = \frac{1}{\lambda}, \qquad \|b_t\|^2_{V_{t-1}^{-1}} = \frac{L^2}{\lambda}, \qquad \|g\|^2_{V_{t-1}^{-1}} = \frac{1}{\lambda}.$$

The key point is that although the observed rewards on $b_1, \dots, b_{t-1}$ contain noise, the estimate

$$\hat{\theta}_{t-1} = V_{t-1}^{-1} \sum_{s=1}^{t-1} b_s X_s$$

lies entirely in $\mathrm{span}\{e_2, \dots, e_t\}$. Therefore,

$$g^\top \hat{\theta}_{t-1} = 0, \qquad b_t^\top \hat{\theta}_{t-1} = 0,$$

because both $g = e_1$ and the fresh arm $b_t = L e_{t+1}$ are orthogonal to $\mathrm{span}\{e_2, \dots, e_t\}$.

Thus, even in the stochastic-reward setting, the comparison between g and b_t is determined purely by the confidence terms:

$$\mathrm{UCB}_t(g) = \frac{\beta}{\sqrt{\lambda}}, \qquad \mathrm{UCB}_t(b_t) = \frac{\beta L}{\sqrt{\lambda}}.$$

Since $L > 1$, we have $\mathrm{UCB}_t(b_t) > \mathrm{UCB}_t(g)$ for all $t \le n$. Hence the isotropically initialized algorithm selects b_t at every round and, like π_{const}, incurs cumulative pseudo-regret n. This proves part (2) of Proposition 1.

Algorithm 3: LinUCB with an anisotropic warm start. We now define an initialization that is realized by exactly the trajectory π_{const} would produce. Fix an integer $m \geq 1$, and let

$$V_0' = \lambda I_d + mL^2 \sum_{j=2}^{n+1} e_j e_j^\top, \qquad \hat{\theta}_0' = 0.$$

This corresponds to having observed each bad direction Le_j, $j = 2, \ldots, n+1$, exactly m times before the online interaction begins — precisely the kind of data π_{const} generates if rolled out for mn rounds on a sequence of action sets containing each bad direction. Thus V_0' is a genuine design matrix obtainable from a collection of outer products produced by a constant logging policy.

At $t = 1$,

$$\|g\|^2_{(V_0')^{-1}} = \frac{1}{\lambda}, \qquad \|b_1\|^2_{(V_0')^{-1}} = L^2 \cdot \frac{1}{\lambda + mL^2} = \frac{L^2}{\lambda + mL^2}.$$

Hence, if

$$m > \frac{\lambda(L^2 - 1)}{L^2},$$

then $\frac{1}{\lambda} > \frac{L^2}{\lambda + mL^2}$, and therefore

$$\text{UCB}_1'(g) > \text{UCB}_1'(b_1).$$

So the warm-started algorithm selects g at round 1.

Suppose now that the algorithm continues to select g for the first s rounds. Then

$$V_s' = \lambda I_d + mL^2 \sum_{j=2}^{n+1} e_j e_j^\top + s\, e_1 e_1^\top.$$

Since each reward observed on g has mean 1,

$$\sum_{i=1}^{s} X_i = s + \sum_{i=1}^{s} \eta_i,$$

and therefore

$$g^\top \hat{\theta}_s' = e_1^\top (V_s')^{-1} \sum_{i=1}^{s} g X_i = \frac{1}{\lambda + s} \sum_{i=1}^{s} X_i.$$

Applying Lemma 4 with $\mu = 1$, we obtain that with probability at least $1 - \delta$, simultaneously for all $1 \leq s \leq n$,

$$\frac{1}{s} \sum_{i=1}^{s} X_i \geq 1 - \sqrt{\frac{2\sigma^2 \log(2n/\delta)}{s}}.$$

On this event,

$$g^\top \hat{\theta}_s' \geq \frac{s}{\lambda + s} \left(1 - \sqrt{\frac{2\sigma^2 \log(2n/\delta)}{s}}\right).$$

For every fresh bad arm $b_{s+1} = Le_{s+2}$,

$$b_{s+1}^\top \hat{\theta}_s' = 0,$$

since the online trajectory has only sampled the direction e_1, and the warm-start contributes only to the design matrix, not to the response vector. Moreover,

$$\|b_{s+1}\|^2_{(V_s')^{-1}} = \frac{L^2}{\lambda + mL^2}, \qquad \|g\|^2_{(V_s')^{-1}} = \frac{1}{\lambda + s}.$$

On the same high-probability event,

$$\text{UCB}_{s+1}'(g) \geq \frac{s}{\lambda + s} \left(1 - \sqrt{\frac{2\sigma^2 \log(2n/\delta)}{s}}\right) + \frac{\beta}{\sqrt{\lambda + s}},$$

while

$$\mathrm{UCB}'_{s+1}(b_{s+1}) = \frac{\beta L}{\sqrt{\lambda + mL^2}}.$$

Consequently, if m is chosen large enough so that for every $1 \le s \le n$,

$$\frac{s}{\lambda + s}\left(1 - \sqrt{\frac{2\sigma^2 \log(2n/\delta)}{s}}\right) + \frac{\beta}{\sqrt{\lambda + s}} > \frac{\beta L}{\sqrt{\lambda + mL^2}}, \tag{7}$$

then, with probability at least $1 - \delta$, the warm-started algorithm selects g at every round, and the cumulative pseudo-regret over the n online rounds is $O(1)$. This proves part (3) of Proposition 1; condition (7) is the explicit "m sufficiently large" requirement referenced in the proposition statement. □

B.2 Proof of Lemma 5 (pseudo-orthogonal three-way separation)

Lemma 5 (Pseudo-orthogonal three-way separation; *negative result*)**.** *The conclusions of Proposition 1 continue to hold when the strict orthogonality of $\{b_t\}_{t=1}^n \cup \{g\}$ is replaced by $\bar{\mu}$-pseudo-orthogonality (all pairwise inner products bounded by $\bar{\mu}$ in absolute value), provided*

$$\bar{\mu}^2 \;\le\; \frac{(L-1)^2\,\lambda}{16\,L^4\,n}. \tag{8}$$

The threshold scaling $\bar{\mu} \asymp 1/\sqrt{n}$ is sharp: at this scale the cumulative cross-direction information accumulated over n rounds becomes comparable to the regularization, and the orthogonality argument breaks down.

Proof sketch. We modify the construction of Appendix B.1 as follows. Replace the strict orthogonality of $\{b_t\}_{t=1}^n \cup \{g\}$ with $\bar{\mu}$-pseudo-orthogonality, i.e., $|\langle b_s, b_t\rangle| \le \bar{\mu}$ for $s \ne t$ and $|\langle b_t, g\rangle| \le \bar{\mu}$ for all t. We track perturbations of three quantities through the original argument: (i) $\|g\|^2_{V_t^{-1}}$ in the cold-start phase; (ii) $\|b_t\|^2_{V_{t-1}^{-1}}$ in the cold-start phase; (iii) $g^\top \widehat{\theta}_{t-1}$ in the cold-start phase.

(i) Perturbation of $\|g\|^2_{V_t^{-1}}$. By Sherman–Morrison applied to $V_t = V_{t-1} + b_t b_t^\top$,

$$\|g\|^2_{V_t^{-1}} - \|g\|^2_{V_{t-1}^{-1}} \;=\; -\frac{\langle b_t, V_{t-1}^{-1} g\rangle^2}{1 + \|b_t\|^2_{V_{t-1}^{-1}}}.$$

In the strict-orthogonality case, $\langle b_t, V_{t-1}^{-1} g\rangle = 0$; under pseudo-orthogonality, induction shows $|\langle b_t, V_{t-1}^{-1} g\rangle| \le L\bar{\mu}/\lambda + O(\bar{\mu}^2 t/\lambda^2)$, so the per-step perturbation of $\|g\|^2_{V_t^{-1}}$ is at most $L^2\bar{\mu}^2/\lambda^2$. Cumulating over $t \le n$,

$$\left|\|g\|^2_{V_n^{-1}} - 1/\lambda\right| \;\le\; \frac{L^2\bar{\mu}^2 n}{\lambda^2}.$$

(ii) Perturbation of $\|b_t\|^2_{V_{t-1}^{-1}}$. The same argument applied to the fresh direction b_t, taking into account that b_t has off-diagonal cosines bounded by $\bar{\mu}$ with all earlier b_s, gives

$$\left|\|b_t\|^2_{V_{t-1}^{-1}} - L^2/\lambda\right| \;\le\; \frac{L^4\bar{\mu}^2 n}{\lambda^2}.$$

(iii) Perturbation of $g^\top \widehat{\theta}_{t-1}$. Using the explicit formula $\widehat{\theta}_{t-1} = V_{t-1}^{-1} \sum_{s<t} b_s X_s$ and the sub-Gaussian noise bound,

$$|g^\top \widehat{\theta}_{t-1}| \;\le\; \frac{L\bar{\mu}\sigma\sqrt{n\log(1/\delta)}}{\lambda + L^2} + O\big(\tfrac{\bar{\mu}^2 n}{\lambda}\big),$$

which is $o(1)$ under condition (8).

Cold-start ordering preserved. The cold-start LinUCB ordering $\mathrm{UCB}_t(b_t) > \mathrm{UCB}_t(g)$ is, in the strict-orthogonal case, the inequality $L\sqrt{1/\lambda} > \sqrt{1/\lambda}$ which holds with margin $(L-1)\sqrt{1/\lambda}$. The

cumulative perturbation analysis above shows that under condition (8), the perturbation of either side of this inequality is at most $(L^2\bar{\mu}^2 n/\lambda^2)\cdot\sqrt{\lambda}/2 = L^2\bar{\mu}^2 n/(2\lambda^{3/2})$, which under condition (8) is $\le (L-1)/(4L)\sqrt{1/\lambda}$, preserving the strict ordering.

Warm-start ordering preserved. In the warm-start phase, the design matrix $V_0' = \lambda I + mL^2\sum_{t=1}^{n} b_t b_t^\top$ has eigenvalues $\lambda + mL^2(1 \pm O(\bar{\mu}^2 n))$ on $\mathrm{span}\{b_t\}$ and λ on the orthogonal complement (modulo lower-order $\bar{\mu}^2$-terms). The same Sherman–Morrison-style analysis applied to the warm-start phase gives that $\|g\|^2_{(V_0')^{-1}}$ remains close to $1/\lambda$ and $\|b_t\|^2_{(V_0')^{-1}}$ remains close to $L^2/(\lambda + mL^2)$, with all errors of order $\bar{\mu}^2 n$. Condition (8) ensures these errors are below the gap that drives the cold-vs-warm ordering, so the warm-started LinUCB still selects g at every round under the same explicit "m sufficiently large" condition as in the strict-orthogonal proof.

Sharpness of $\bar{\mu} \asymp 1/\sqrt{n}$. If $\bar{\mu}^2 n = \Omega(\lambda)$, the cumulative error in $\|g\|^2_{V_n^{-1}}$ is $\Omega(1)$, comparable to the regularization term itself, and the orthogonality argument breaks down: cold-start LinUCB collects enough cross-direction information to estimate $\theta^\star$'s component on g, and the linear-regret behaviour fails. □

B.3 Proof of Lemma 2 (coherent features)

Proof. By Sherman–Morrison, the rank-1 update $V_{t+1} = V_t + x_{t,a}x_{t,a}^\top$ gives

$$V_{t+1}^{-1} \;=\; V_t^{-1} - \frac{V_t^{-1}x_{t,a}x_{t,a}^\top V_t^{-1}}{1 + \|x_{t,a}\|^2_{V_t^{-1}}}.$$

Substituting $u = V_t^{-1/2}x_{t,a_t^\star}$, $v = V_t^{-1/2}x_{t,a}$,

$$\|x_{t,a_t^\star}\|^2_{V_{t+1}^{-1}} \;=\; \|u\|_2^2 - \frac{\langle u, v\rangle^2}{1 + \|v\|_2^2}.$$

The lower-coherence assumption $\bar{\alpha} \in (0,1]$ implies $\langle x_{t,a_t^\star}, x_{t,a}\rangle \ge \bar{\alpha}\|x_{t,a_t^\star}\|\|x_{t,a}\| > 0$. A standard quadratic-form change-of-variables shows that V_t^{-1}, being symmetric positive definite, distorts cosines by at most a factor $\kappa(V_t)^{1/2}$, the square root of its condition number. We can absorb this distortion into a smaller positive constant $\bar{\alpha}' \in (0, \bar{\alpha}]$ depending on λ, L but *not* on T or d, since V_t has eigenvalues bounded between λ and $\lambda + L^2 t$ on the rank-r active subspace and the ratio of eigenvalues entering this argument is bounded over rounds in which $\|x\|^2_{V_t^{-1}} \ge \Omega(1/t)$ — i.e., over the rounds that contribute non-trivially to the regret. Hence

$$\langle u, v\rangle \;\ge\; \bar{\alpha}'\,\|u\|_2\|v\|_2,$$

which gives

$$\|x_{t,a_t^\star}\|^2_{V_{t+1}^{-1}} \;\le\; \|u\|_2^2\Big(1 - \frac{(\bar{\alpha}')^2\|v\|_2^2}{1 + \|v\|_2^2}\Big) \;=\; (1 - (\bar{\alpha}')^2\rho_t)\,\|x_{t,a_t^\star}\|^2_{V_t^{-1}}.$$

The recursive bound implies that after k pulls of any sub-optimal action whose feature V_t^{-1}-norm is bounded above, $\|x_{t,a_t^\star}\|^2_{V_t^{-1}}$ contracts by a factor $(1 - c\bar{\alpha}^2)^k$ for an absolute constant c. Hence the standard elliptical-potential argument [1, Lemma 11] now produces a determinant-ratio bound with an extra factor of $1/\bar{\alpha}^2$, yielding $R_T^{\mathrm{cold}} = \widetilde{O}(\beta_T\sqrt{rT}/\bar{\alpha})$. The constants depend only on $\bar{\alpha}, \lambda, L$, not on the ambient dimension d.

The non-existence claim follows by contradiction: if a linear-regret instance existed with $\bar{\alpha} > 0$, the multiplicative shrinkage above would deliver $R_T^{\mathrm{cold}} = \widetilde{O}(\sqrt{T})$, contradicting the linear lower bound. □

B.4 Proof of Lemma 1 (saturated coverage)

Proof. Cold-start LinUCB obeys the standard regret bound (4) which in the projected setting (where features lie in $\mathrm{Im}(P)$) reduces to $R_T^{\mathrm{cold}} \le C\beta_T\sqrt{rT\log(1 + L^2T/\lambda)}$. Under the hypothesis $r\log T \le c\sqrt{T}$ of Lemma 1, $\beta_T = O(\sigma\sqrt{r\log T} + \sqrt{\lambda}S) = O(T^{1/4} + 1)$, so

$R_T^{\mathrm{cold}} = \widetilde{O}(\beta_T\sqrt{rT}) = \widetilde{O}(T^{1/4} \cdot \sqrt{rT})$. Since $r \le c\sqrt{T}/\log T$, this gives $R_T^{\mathrm{cold}} = \widetilde{O}(T^{1/4} \cdot T^{3/4}/\sqrt{\log T}) = \widetilde{O}(T/\sqrt{\log T})$, which is sublinear by a logarithmic factor; in the asymptotic average-reward sense, this is $o(T)$ as $T \to \infty$. (The phrase "$R_T^{\mathrm{cold}} = \widetilde{O}(\sqrt{rT})$" in the lemma statement is the standard LinUCB shorthand that absorbs β_T into the polylogarithmic $\widetilde{O}$; we make this explicit here for clarity.)

The meta-algorithm's bound (3) differs from the cold-start bound only in (i) the $\sqrt{(T-\tau)/T}$ prefactor, (ii) the $\log\big(1 + L^2(T-\tau)/(\lambda+\nu\tau)\big)$ factor inside the square root, and (iii) the additive $\tau\varepsilon_0$ term. Direct comparison gives $R_T^{\mathrm{cold}} - R_T^{\mathrm{meta}} + \tau\varepsilon_0 \le (1 - \Psi(\tau,\nu,T))R_T^{\mathrm{cold}}$, which is the stated bound.

For this gap to exceed a constant in T, we need $(1-\Psi)R_T^{\mathrm{cold}} = \omega(1)$. Bounding Ψ from below by $\sqrt{\lambda/(\lambda+\nu\tau)}$ and using $R_T^{\mathrm{cold}} = \widetilde{O}(\sqrt{T})$ shows this requires $\nu\tau = \Omega(R_T^{\mathrm{cold}}/\Psi)$, hence $\nu\tau = \Theta(\sqrt{T})$, hence $\tau = \Theta(\sqrt{T}/\nu)$. For this τ, the additive warm-start cost is $\tau\varepsilon_0 = \Theta(\sqrt{T}\varepsilon_0/\nu)$, which is comparable to the alleged improvement, so the meta-algorithm is no longer better than cold-start by more than a constant factor. □

B.5 Proof of Lemma 6 (sparse reward in dense features)

Lemma 6 (Sparse reward in dense features; *negative result*)**.** *Fix T, r, d with $r \ll d$ and $r = T$. There exists a contextual bandit instance with $\theta^\star$ supported on a rank-r subspace $\mathrm{Im}(P)$, on which cold-start LinUCB suffers regret $R_T^{\mathrm{cold}} = \Theta(T)$, while the meta-algorithm of Section 3.2, warm-started from $\widetilde{\Omega}(r)$ offline rollouts of an oracle that restricts attention to $\mathrm{Im}(P)$, attains $R_T^{\mathrm{meta}} \le \tau\varepsilon_0 + O(1)$. In particular, when the oracle is exact ($\varepsilon_0 = 0$), this gives $R_T^{\mathrm{meta}} = O(1)$, and the ratio $R_T^{\mathrm{cold}}/R_T^{\mathrm{meta}}$ diverges as $T \to \infty$.*

Proof. We give an explicit construction by lifting Proposition 1 from a low-dimensional space into a high-dimensional ambient space, then verifying the regret bounds for cold-start LinUCB and for the warm-started meta-algorithm.

Construction. Fix horizon $T = n$ and effective dimension $r = n + 1$. Embed $\mathbb{R}^{n+1}$ as a fixed r-dimensional subspace $\mathrm{Im}(P) \subseteq \mathbb{R}^d$ (with $d \gg r$ arbitrary), via an orthonormal frame $u_0, u_1, \ldots, u_n \in \mathbb{R}^d$ with $P = \sum_{j=0}^n u_j u_j^\top$. Set

$$\theta^\star = u_0 \in \mathrm{Im}(P), \qquad \|\theta^\star\|_2 = 1.$$

At round $t \in [T]$, the action set has two arms,

$$g_t \;=\; u_0, \qquad b_t \;=\; L\,u_t, \qquad L > 1,$$

both lying in $\mathrm{Im}(P)$ (so Assumption B is satisfied with projector P). By construction:

- $\|g_t\|_2 = 1$, $\|b_t\|_2 = L$.
- $\langle\theta^\star, g_t\rangle = 1$ and $\langle\theta^\star, b_t\rangle = 0$, so g_t is the optimal arm with constant gap 1.
- Both arms lie in $\mathrm{Im}(P)$ and have no component in $\mathrm{Im}(P^\perp)$, so the determinant ratio of Lemma 9 factorizes cleanly across the subspaces.

Cold-start lower bound. This construction is exactly Proposition 1 embedded in $\mathbb{R}^d$; the same argument as in Appendix B.1 shows that cold-start LinUCB with $V_0 = \lambda I_d$ pulls b_t at every round and incurs

$$R_T^{\mathrm{cold}} \;=\; T \;=\; \Theta(T).$$

Warm-start upper bound. Suppose the meta-algorithm of Section 3.2 is warm-started by $N = \widetilde{\Omega}(r)$ offline rollouts of an oracle that pulls each b_t, $t = 1, \ldots, n$, exactly $m \ge 1$ times. The resulting offline empirical Gram matrix has the block form

$$V^{(i^\star)} \;=\; \lambda I_d + mL^2 \sum_{t=1}^n u_t u_t^\top,$$

which satisfies the spectral lower bound $V^{(i^\star)} \succeq (\lambda + mL^2)P_{\setminus 0}$ on the n-dim subspace $\mathrm{span}\{u_1, \ldots, u_n\}$, where $P_{\setminus 0}$ is the orthogonal projector onto that subspace. Assumption B is satisfied with rate $\nu = mL^2/\tau$ for $\tau \geq N$.

The argument of Appendix B.1 (Algorithm 3 case) shows that for any m satisfying $m > \lambda(L^2-1)/L^2$, the warm-started LinUCB selects g_t at every round t with probability at least $1-\delta$. The cumulative regret over the T online rounds is then $O(1)$, so $R_T^{\mathrm{meta}} \leq \tau\varepsilon_0 + O(1)$.

Ratio. Under Assumption A with $\varepsilon_0 = 0$ (the oracle is exact, since the b_t's have $\langle \theta^\star, b_t \rangle = 0$ and the routing phase only ever pulls b_t),

$$\frac{R_T^{\mathrm{cold}}}{R_T^{\mathrm{meta}}} \;=\; \frac{\Theta(T)}{O(1)} \;=\; \Theta(T) \to \infty.$$

The separation is divergent as $T \to \infty$, demonstrating the warm-start-essential corner. □

B.6 Proof of Lemma 7 (vanishing gap)

Lemma 7 (Vanishing gap: warm-start can be essential; *negative result*)**.** *There exist contextual bandit instances of effective dimension $r \ll T$ on which the per-round suboptimality gap of a uniformly positive fraction of rounds is below any $\eta > 0$ (so no $\Delta_{\min} > 0$ uniformly bounds the gaps), cold-start LinUCB suffers regret $\widetilde{\Omega}(\sqrt{T})$, and the meta-algorithm of Section 3.2 with warm-start length τ satisfying the coverage condition (Assumption B) achieves $\tau\varepsilon_0 + \widetilde{O}(\Psi(\tau,\nu,T)\sqrt{T})$ where $\Psi = \sqrt{\lambda/(\lambda+\nu\tau)}$ can be made arbitrarily small.*

Proof. We construct a 2-dimensional instance with vanishing gaps, verify the cold-start lower bound, and then bound the warm-started regret.

Construction. Set $d = r = 2$, $\theta^\star = e_1 \in \mathbb{R}^2$, and $\lambda > 0$. At each round $t \in [T]$, sample a context $u_t \sim \mathrm{Unif}[0,1]$ independently, and present the two-action set $\mathcal{A}_t = \{e_1, \quad u_t e_1 + \sqrt{1-u_t^2}\, e_2\}$. Each feature has $\|x_{t,a}\|_2 = 1$, so $L = 1$, and the rewards are $\mu_t(\text{first}) = 1$ and $\mu_t(\text{second}) = u_t$. The optimal action is the first arm with instantaneous gap $\mu_t(a_t^\star) - \mu_t(\text{second}) = 1 - u_t \sim \mathrm{Unif}[0,1]$.

Vanishing-gap property. For every $\eta \in (0,1)$, the fraction of rounds with action gap below η converges in probability to $\Pr(1-u_t \leq \eta) = \eta$, so *no* positive lower bound on the gap holds uniformly. The conclusion of Lemma 3 cannot apply.

Cold-start lower bound. On this $d = 2$ stochastic linear bandit instance with random feature matrix and Bernoulli-bounded rewards, the standard $\Omega(\sqrt{T})$ minimax lower bound for stochastic linear bandits with $d = 2$ applies [27, Theorem 24.1]: for any algorithm there exists a parameter $\theta^\star \in \mathbb{R}^2$ with $\|\theta^\star\|_2 \leq 1$ on which the algorithm suffers regret $\Omega(\sqrt{T})$. The construction here realizes such an instance because the second arm's feature direction varies over the unit circle as u_t ranges, and rewards near the boundary $u_t \approx 1$ provide vanishingly small information per pull. Hence

$$R_T^{\mathrm{cold}} \;=\; \widetilde{\Omega}(\sqrt{T}).$$

Warm-started upper bound. Consider warm-started LinUCB with $V_0 = \lambda I_2 + \nu\tau P$, where $P = I_2$ is the identity on the (rank-2) reward-relevant subspace. The cold-start argument (10)–(11) now uses $V_t \succeq V_0$ throughout, so

$$\sum_{t=1}^{T} \|x_{t,A_t}\|^2_{V_t^{-1}} \;\leq\; 2\log\frac{\det V_{T+1}}{\det V_0} \;\leq\; 2r\log\Big(1 + \tfrac{T}{\lambda+\nu\tau}\Big).$$

By the per-round confidence bound (9) and Cauchy–Schwarz over $t \in [T]$,

$$R_T^{\mathrm{warm}} \;\leq\; 2\beta_T\sqrt{T\sum_{t=1}^{T}\|x_{t,A_t}\|^2_{V_t^{-1}}} \;\leq\; 2\beta_T\sqrt{2rT\log\Big(1+\tfrac{T}{\lambda+\nu\tau}\Big)} \;=\; \widetilde{O}(\Psi(\tau,\nu,T)\sqrt{T}),$$

where $\Psi(\tau, \nu, T) \asymp \sqrt{\lambda/(\lambda + \nu\tau)}$ for $\nu\tau \gg \lambda$. The meta-algorithm of Section 3.2 run with expert coverage rate ν on the first τ rounds (Assumption B) yields exactly this design matrix at the start of the LinUCB phase, plus an additive $\tau\varepsilon_0$ term from the warm-start phase (Lemma 8), establishing the bound $R_T^{\text{meta}} \le \tau\varepsilon_0 + \widetilde{O}(\Psi\sqrt{T})$ claimed by the lemma.

Polynomial separation. Choosing $\nu\tau = \lambda T^\alpha$ for any $\alpha \in (0, 1)$ gives $\Psi = T^{-\alpha/2}$, hence $R_T^{\text{meta}} - \tau\varepsilon_0 = \widetilde{O}(T^{1/2-\alpha/2})$ while $R_T^{\text{cold}} = \widetilde{\Omega}(T^{1/2})$, so the ratio $R_T^{\text{cold}}/R_T^{\text{meta}} = \widetilde{\Omega}(T^{\alpha/2})$ diverges. This is the warm-start-essential conclusion of the lemma. □

B.7 Proof of Lemma 3 (uniform gap)

Proof. We give a self-contained proof of the gap-dependent regret bound for cold-start LinUCB. The argument has four steps.

Step 1: per-round confidence bound. On the standard self-normalized concentration event of **(author?)** [1], which holds with probability at least $1 - \delta$, the LinUCB estimate $\widehat{\theta}_t = V_t^{-1} b_t$ satisfies

$$\big\|\widehat{\theta}_t - \theta^\star\big\|_{V_t} \;\le\; \beta_T \quad \text{for all } t \ge 1.$$

Combined with the LinUCB selection rule $A_t \in \arg\max_a \langle \widehat{\theta}_t, x_{t,a}\rangle + \beta_T \|x_{t,a}\|_{V_t^{-1}}$, this yields the per-round upper-confidence bound

$$\mu_t(a_t^\star) - \mu_t(A_t) \;\le\; 2\beta_T \|x_{t,A_t}\|_{V_t^{-1}}. \tag{9}$$

Step 2: bad-pull characterization. On any round t with $A_t \ne a_t^\star$, the uniform-gap hypothesis of Lemma 3 (for every t and every $a \in \mathcal{A}_t \setminus \{a_t^\star\}$, $\mu_t(a_t^\star) - \mu_t(a) \ge \Delta_{\min}$) gives $\mu_t(a_t^\star) - \mu_t(A_t) \ge \Delta_{\min}$, so combined with (9),

$$\|x_{t,A_t}\|_{V_t^{-1}} \;\ge\; \frac{\Delta_{\min}}{2\beta_T}.$$

Equivalently,

$$\mathbf{1}[A_t \ne a_t^\star] \;\le\; \big(2\beta_T/\Delta_{\min}\big)^2 \, \|x_{t,A_t}\|^2_{V_t^{-1}}. \tag{10}$$

Step 3: elliptical potential bound. By the elliptical potential lemma [1, Lemma 11],

$$\sum_{t=1}^{T} \min\big(\|x_{t,A_t}\|^2_{V_t^{-1}},\, 1\big) \;\le\; 2\log\frac{\det V_{T+1}}{\det V_1} \;\le\; 2r\log\Big(1 + \tfrac{L^2 T}{\lambda}\Big), \tag{11}$$

where the second inequality uses $\det V_{T+1} \le \big(\frac{\operatorname{tr}(V_{T+1})}{r}\big)^r$ and $\operatorname{tr}(V_{T+1}) \le r\lambda + L^2 T$ (since features lie in a rank-r subspace by hypothesis). The truncation by 1 in (11) is immaterial: the standard regularization $V_1 = \lambda I$ together with $\|x_{t,a}\|_2 \le L$ gives $\|x_{t,A_t}\|^2_{V_t^{-1}} \le L^2/\lambda$, so the truncated and untruncated sums differ by at most a constant absorbed into C.

Step 4: counting bad pulls and bounding regret. Let $N_T := |\{t \in [T] : A_t \ne a_t^\star\}|$ be the number of suboptimal pulls. Combining (10) (summed over t) with (11),

$$N_T \;=\; \sum_{t=1}^{T} \mathbf{1}[A_t \ne a_t^\star] \;\le\; \big(2\beta_T/\Delta_{\min}\big)^2 \sum_{t=1}^{T} \|x_{t,A_t}\|^2_{V_t^{-1}} \;\le\; \frac{8r\beta_T^2}{\Delta_{\min}^2}\log\Big(1 + \tfrac{L^2 T}{\lambda}\Big).$$

The cumulative regret is bounded by summing the actual instantaneous gaps, which are themselves bounded above by $2\beta_T \|x_{t,A_t}\|_{V_t^{-1}}$ on every round (whether or not it is a bad-pull round, since

$\|x_{t,A_t}\|_{V_t^{-1}} \geq 0$ trivially):

$$\begin{aligned} R_T^{\text{cold}} &= \sum_{t=1}^{T} \big(\mu_t(a_t^\star) - \mu_t(A_t)\big)\, \mathbf{1}[A_t \neq a_t^\star] \\ &\leq \sum_{t=1}^{T} 2\beta_T \, \|x_{t,A_t}\|_{V_t^{-1}} \, \mathbf{1}[A_t \neq a_t^\star] \quad \text{(by (9))} \\ &\leq \frac{4\beta_T^2}{\Delta_{\min}} \sum_{t=1}^{T} \|x_{t,A_t}\|^2_{V_t^{-1}} \quad \text{(by (10): } \Delta_{\min} \leq 2\beta_T \|x\|_{V_t^{-1}} \text{ on bad rounds)} \\ &\leq \frac{8r\beta_T^2}{\Delta_{\min}} \log\Big(1 + \tfrac{L^2 T}{\lambda}\Big) \;=\; \widetilde{O}\Big(\frac{r^2 \log^2 T}{\Delta_{\min}}\Big), \end{aligned}$$

using $\beta_T = O(\sigma\sqrt{r \log T} + \sqrt{\lambda} S)$ in the last step. This is the stated cold-start bound.

Warm-started LinUCB attains the same order. For any warm-started LinUCB initialized with $V_0 = \lambda I + M$ where $M \succeq 0$ has $\operatorname{tr}(M) \leq L^2 N$ for some $N \geq 0$ (corresponding to at most N offline rollouts), the same argument applies with the substitution $V_t \mapsto V_t$ throughout (the warm-start only enlarges V_t, so all confidence widths $\|x\|_{V_t^{-1}}$ shrink). The determinant ratio in (11) becomes $\log(\det V_{T+1} / \det V_1) \leq r \log\big(1 + L^2(T+N)/(r\lambda)\big)$, which is at most a constant factor larger than the cold-start ratio when $N = O(T)$. Hence warm-started LinUCB attains the same $\widetilde{O}(r^2 \log^2 T / \Delta_{\min})$ rate, and no warm-start improves the order of cold-start regret on instances with uniform gap. □

B.8 Proof of Theorem 1

We prove Theorem 1 via two self-contained lemmas: a warm-start phase contribution (Lemma 8) and a post-handover phase contribution (Lemma 9) stated in "determinant-ratio" form. The determinant-ratio form is reused without modification in Appendix B.9 for Theorem 2.

Convention. Throughout the proofs of Lemmas 8 and 9, we adopt the notational convention $\mathcal{T}_{\text{exp}} = \{1, \ldots, \tau\}$ and $\mathcal{T}_{\text{lin}} = \{\tau + 1, \ldots, T\}$. This is without loss of generality: the elliptical-potential argument used in the post-handover lemma applies to any sequence of LinUCB-routed rounds whose design matrix at the start of the LinUCB phase satisfies the spectral lower bound $V \succeq (\lambda + \nu\tau)P$ from Assumption B. Equivalently, the bound is monotone in the order of rounds: relabeling expert rounds to come first only weakens the bound on the LinUCB covariance and therefore strengthens the regret bound. We use the contiguous form for notational simplicity; the partition-form statements in the main text follow by applying the contiguous bound to the worst-case ordering.

Lemma 8 (Warm-start phase contribution)**.** *Under Assumption A, the cumulative pseudo-regret of the meta-algorithm on the expert-routed rounds satisfies*

$$\sum_{t \in \mathcal{T}_{\text{exp}}} \big(\mu_t(a_t^\star) - \mu_t(A_t)\big) \;\leq\; \tau\varepsilon_0.$$

Proof of Lemma 8. For each $t \in \mathcal{T}_{\text{exp}}$, by definition the meta-algorithm plays the action proposed by some frozen expert $I_t \neq 0$, i.e., $A_t = a_t^{(I_t)}$. Assumption A gives $\mu_t(a_t^\star) - \mu_t(a_t^{(I_t)}) = \Delta_t^{(I_t)} \leq \varepsilon_0$ for every such t. Summing over the $\tau = |\mathcal{T}_{\text{exp}}|$ rounds yields the stated bound; note that the rounds need not be contiguous, only that they share the cardinality τ. □

Lemma 9 (Post-handover regret in determinant-ratio form)**.** *Under Assumption B and the linear reward model, on the standard self-normalized concentration event (which has probability at least $1 - \delta$), for every $\tau \in \{1, \ldots, T-1\}$,*

$$\sum_{t=\tau+1}^{T} \big(\mu_t(a_t^\star) - \mu_t(A_t)\big) \;\leq\; 2\beta_T \sqrt{2(T-\tau)\, \log \frac{\det(V_{T+1})}{\det(V_{\tau+1})}}. \tag{12}$$

Moreover, the determinant ratio admits two upper bounds: the anisotropic *bound*

$$\log \frac{\det(V_{T+1})}{\det(V_{\tau+1})} \;\le\; \sum_{j=1}^{r} \log\Big(1 + \tfrac{L^2(T-\tau)}{\sigma_j(V_{\tau+1})}\Big) \tag{13}$$

in terms of the eigenvalues $\sigma_j(V_{\tau+1})$ of $V_{\tau+1}$ on $\mathrm{Im}(P)$, and the isotropic *relaxation under Assumption B,*

$$\log \frac{\det(V_{T+1})}{\det(V_{\tau+1})} \;\le\; r \log\Big(1 + \tfrac{L^2(T-\tau)}{\lambda+\nu\tau}\Big). \tag{14}$$

Proof of Lemma 9. The proof has three steps: a per-round confidence bound, a Cauchy–Schwarz aggregation, and an elliptical-potential bound on the determinant ratio.

Step 1: per-round confidence bound. On the standard self-normalized concentration event of [1] (their Theorem 2), the LinUCB estimate $\widehat{\theta}_t = V_t^{-1} b_t$ satisfies

$$\|\widehat{\theta}_t - \theta^\star\|_{V_t} \;\le\; \beta_T \quad \text{for all } t \ge 1,$$

with probability at least $1-\delta$. Combining this with the LinUCB selection rule $A_t \in \arg\max_a \langle \widehat{\theta}_t, x_{t,a}\rangle + \beta_T \|x_{t,a}\|_{V_t^{-1}}$ gives the well-known per-round upper-confidence bound for $t > \tau$:

$$\mu_t(a_t^\star) - \mu_t(A_t) \;\le\; 2\beta_T \|x_{t,A_t}\|_{V_t^{-1}}. \tag{15}$$

Step 2: Cauchy–Schwarz aggregation. Squaring and applying Cauchy–Schwarz over $t = \tau+1, \dots, T$ gives

$$\sum_{t=\tau+1}^{T} \mu_t(a_t^\star) - \mu_t(A_t) \;\le\; 2\beta_T \sqrt{(T-\tau) \sum_{t=\tau+1}^{T} \|x_{t,A_t}\|^2_{V_t^{-1}}}. \tag{16}$$

Step 3: elliptical potential bound. By the elliptical potential lemma [1, Lemma 11],

$$\sum_{t=\tau+1}^{T} \min\big(\|x_{t,A_t}\|^2_{V_t^{-1}}, 1\big) \;\le\; 2 \log \frac{\det V_{T+1}}{\det V_{\tau+1}};$$

since $\|x_{t,A_t}\|^2_{V_t^{-1}} \le L^2/\lambda_{\min}(V_{\tau+1}) \le L^2/\lambda$, the truncation can be absorbed into the universal constant C in (3), so

$$\sum_{t=\tau+1}^{T} \|x_{t,A_t}\|^2_{V_t^{-1}} \;\le\; 2 \log \frac{\det V_{T+1}}{\det V_{\tau+1}}. \tag{17}$$

Combining (16) and (17) establishes (12).

Determinant ratio: anisotropic bound (13). Let $\{u_j\}_{j=1}^r$ be an orthonormal basis of $\mathrm{Im}(P)$ aligned with the eigenvectors of $V_{\tau+1}|_{\mathrm{Im}(P)}$, with eigenvalues $\sigma_1(V_{\tau+1}) \ge \cdots \ge \sigma_r(V_{\tau+1})$, and complete to an orthonormal basis of $\mathbb{R}^d$ by adding a basis of $\mathrm{Im}(P)^\perp$. By Assumption B, all action vectors $x_{t,a}$ lie in $\mathrm{Im}(P)$, so $V_T - V_{\tau+1} = \sum_{t=\tau+1}^{T} x_{t,A_t} x_{t,A_t}^\top$ has support contained in $\mathrm{Im}(P)$. Therefore the determinant ratio factorizes across the two subspaces, and only the r-dimensional reward-relevant block contributes:

$$\frac{\det V_{T+1}}{\det V_{\tau+1}} \;=\; \frac{\det\big(V_{\tau+1}|_{\mathrm{Im}(P)} + M\big)}{\det\big(V_{\tau+1}|_{\mathrm{Im}(P)}\big)},$$

where $M = \sum_{t=\tau+1}^{T} (P x_{t,A_t})(P x_{t,A_t})^\top \succeq 0$ is the restriction of the post-handover empirical Gram matrix to $\mathrm{Im}(P)$. Diagonalizing $V_{\tau+1}|_{\mathrm{Im}(P)} = \mathrm{diag}(\sigma_1, \dots, \sigma_r)$ in the u_j-basis and using $\det(D+M) \le \prod_j (D_{jj} + M_{jj})$ (Hadamard's inequality applied to the PSD matrix $D^{-1/2} M D^{-1/2} + I$), together with $M_{jj} = \sum_{t>\tau} (u_j^\top x_{t,A_t})^2 \le L^2(T-\tau)$, gives

$$\log \frac{\det V_{T+1}}{\det V_{\tau+1}} \;\le\; \sum_{j=1}^{r} \log\Big(1 + \tfrac{M_{jj}}{\sigma_j(V_{\tau+1})}\Big) \;\le\; \sum_{j=1}^{r} \log\Big(1 + \tfrac{L^2(T-\tau)}{\sigma_j(V_{\tau+1})}\Big),$$

which is (13).

Determinant ratio: isotropic bound (14). Assumption B gives the worst-case eigenvalue lower bound $\sigma_j(V_{\tau+1}) \ge \lambda + \nu\tau$ for every $j \in \{1, \dots, r\}$. Substituting this bound into (13) yields (14). □

Proof of Theorem 1. We assemble the lemmas above. By Lemma 8, the warm-start phase contributes at most $\tau\varepsilon_0$ to the cumulative regret. By Lemma 9, the post-handover phase contributes at most $2\beta_T\sqrt{2(T-\tau)\log(\det V_{T+1}/\det V_{\tau+1})}$, and by (14) the determinant ratio is at most $r\log\big(1+L^2(T-\tau)/(\lambda+\nu\tau)\big)$. Combining and absorbing $2\sqrt{2}$ into the universal constant C gives (3). Equation (4) is the same argument with $\tau=0$, $V_{\tau+1}=\lambda I_d$, and the trivial projector $P=I_d$ replaced by the rank-r projector inside the elliptical-potential bound (the actions still lie in $\mathrm{Im}(P)$ in the cold-start regime, so only r directions contribute to the determinant ratio). For (5), comparing the post-handover term in (3) to (4) term-by-term,

$$\frac{C\beta_T\sqrt{r(T-\tau)\log(1+L^2(T-\tau)/(\lambda+\nu\tau))}}{C\beta_T\sqrt{rT\log(1+L^2T/\lambda)}} \;=\; \sqrt{\frac{T-\tau}{T}}\cdot\sqrt{\frac{\log(1+L^2(T-\tau)/(\lambda+\nu\tau))}{\log(1+L^2T/\lambda)}} \;=\; \Psi(\tau,\nu,T).$$

For $\nu\tau \gg \lambda$, the logarithmic ratio is bounded by a horizon-independent constant (since both numerator and denominator are of order $\log T$), and the prefactor reduces to $\sqrt{(T-\tau)/T}\le 1$, multiplied by a logarithmic correction that behaves like $\sqrt{\lambda/(\lambda+\nu\tau)}$. This is the asymptotic form quoted in the theorem statement. □

B.9 Proof of Theorem 2

The proof of Theorem 2 has three steps that mirror its two parts plus an intermediate optimization. Step 1 establishes the parametric inequality $T(\bar r_T^{\mathrm{meta}}-\bar r_T^{\mathrm{fix}})\ge\Phi(\tau,T)$ for an arbitrary admissible τ by combining the post-handover regret bound from Theorem 1 with the tail-gap inequality of Assumption D. Step 2 lower-bounds the supremum $\sup_\tau\Phi(\tau,T)$ by exhibiting an explicit choice of τ achieving an explicit dimension-explicit bound (18); this is an intermediate step toward the phase-transition statement and is not exposed in the theorem. Step 3 substitutes the resulting bound into the asymptotic regime $\beta_T=O(\sqrt{r\log T})$ and reads off the dimensional phase transition. The optimization-friendly framing in terms of $\Phi(\tau,T)$ is the main novelty relative to a direct "subtract the bounds" argument: it lets us identify $r=o(\sqrt{T}/\log T)$ as a sharp threshold rather than as a sufficient condition.

We rely on Lemma 9 from Appendix B.8, which gives the post-handover regret bound in determinant-ratio form together with its anisotropic and isotropic specializations. For brevity, write

$$G(\tau,T)\;:=\;C\,\beta_T\sqrt{r(T-\tau)\,\log\Big(1+\tfrac{L^2(T-\tau)}{\lambda+\nu\tau}\Big)}$$

for the isotropic post-handover bound (i.e., the right-hand side of (3) minus its $\tau\varepsilon_0$ term, viewed as a function of τ); on the standard self-normalized concentration event, $\sum_{t=\tau+1}^{T}(\mu_t(a_t^\star)-\mu_t(A_t))\le G(\tau,T)$ by combining (12) and (14).

Step 1: Parametric bound, part (i). By construction of the meta-algorithm, on the warm-start phase $t\le\tau$ we have $A_t=a_t^{(i^\star)}$ and hence $\mu_t(A_t)=\mu_t(a_t^{(i^\star)})$ pointwise. Therefore

$$\sum_{t=1}^{\tau}\mathbb{E}\big[\mu_t(A_t)-\mu_t(a_t^{(i^\star)})\big]\;=\;0.$$

On the post-handover phase, write

$$\mu_t(A_t)-\mu_t(a_t^{(i^\star)})\;=\;\big(\mu_t(a_t^\star)-\mu_t(a_t^{(i^\star)})\big)-\big(\mu_t(a_t^\star)-\mu_t(A_t)\big).$$

Summing over $t=\tau+1,\dots,T$ and taking expectation, we apply Assumption D to the first term (which yields a deterministic lower bound $(T-\tau)\varepsilon_{\mathrm{tail}}$) and decompose the second term into the high-probability regret event and its complement:

$$\mathbb{E}\Big[\sum_{t=\tau+1}^{T}(\mu_t(a_t^\star)-\mu_t(A_t))\Big]\;\le\;G(\tau,T)\cdot(1-\delta)+T\cdot\delta\;\le\;G(\tau,T)+1$$

where the first inequality uses the per-round $|r| \le 1$ bound on the failure event $\overline{\mathcal{E}}$ (with $\Pr(\overline{\mathcal{E}}) \le \delta$), and the second uses $\delta = 1/T$. This gives

$$T\big(\bar{r}_T^{\mathrm{meta}} - \bar{r}_T^{\mathrm{fix}}\big) = \sum_{t=\tau+1}^{T} \mathbb{E}\big[\mu_t(A_t) - \mu_t(a_t^{(i^\star)})\big] \\ \ge (T-\tau)\varepsilon_{\mathrm{tail}} - G(\tau,T) - 1 \;=\; \Phi(\tau,T) - 1,$$

which establishes the parametric bound of Theorem 2(i). The multiplicative form $\bar{r}_T^{\mathrm{meta}}/\bar{r}_T^{\mathrm{fix}} \ge 1 + (\Phi(\tau,T)-1)/(T\bar{r}_T^{\mathrm{fix}})$ follows by dividing by $T\bar{r}_T^{\mathrm{fix}} > 0$. The additive -1 is absorbed into the asymptotic $\Omega(T)$ bound below and into practical reporting (where $\Phi(\tau,T)$ is itself $\Omega(T)$ at the optimum, so the -1 correction is negligible). This completes Step 1.

Step 2: Optimal balance (intermediate step). Step 1 established the parametric inequality $T(\bar{r}_T^{\mathrm{meta}} - \bar{r}_T^{\mathrm{fix}}) \ge \Phi(\tau,T)$ for every admissible τ. Maximizing the right-hand side over $\tau \in \{0,1,\dots,T-1\}$ preserves the bound. We record the following dimension-explicit lower bound, which is needed for Step 3:

$$\sup_{\tau} \Phi(\tau,T) \;\ge\; T\varepsilon_{\mathrm{tail}} - \widetilde{C}\,\beta_T\sqrt{rT\log T}, \tag{18}$$

attained in particular at $\tau = 0$. Two observations suffice:

(a) The trivial choice $\tau = 0$ already attains (18). Substituting $\tau = 0$ into Φ gives

$$\Phi(0,T) \;=\; T\varepsilon_{\mathrm{tail}} \;-\; C\beta_T\sqrt{rT\,\log\!\big(1+\tfrac{L^2T}{\lambda}\big)},$$

which equals the right-hand side of (18) with $\widetilde{C} = C$. Hence $\sup_\tau \Phi(\tau,T) \ge \Phi(0,T)$ proves the lower bound. The warm-start machinery is not needed for the dimensional scaling: it influences only the leading constant.

(b) Heuristic balance: locating the leading-order optimum. We use the derivative as a heuristic to identify the order of magnitude of the optimal τ; the rigorous bound is secured by part (a) above. Differentiating Φ in τ and writing $\ell(\tau) := \log\big(1 + L^2(T-\tau)/(\lambda+\nu\tau)\big)$,

$$\partial_\tau \Phi(\tau,T) \;=\; -\varepsilon_{\mathrm{tail}} \;+\; \frac{C\beta_T\sqrt{r}}{2}\cdot\frac{\ell(\tau) - (T-\tau)\ell'(\tau)}{\sqrt{(T-\tau)\,\ell(\tau)}}.$$

Note that $\Phi(\tau,T)$ is a difference of a linear-in-τ term and a $\sqrt{(T-\tau)\,\ell(\tau)}$ term; since the negated second term is *concave* in τ on the relevant range (the composition of a concave $\sqrt{\cdot}$ with a function that has bounded curvature), Φ is concave on this range and the critical point is a local maximum. Setting $\partial_\tau\Phi = 0$ and dropping the lower-order $(T-\tau)\ell'/\ell$ correction gives the heuristic balance equation

$$\varepsilon_{\mathrm{tail}} \;\asymp\; \frac{C\beta_T}{2}\sqrt{\frac{r\log(1+L^2(T-\tau^\star)/(\lambda+\nu\tau^\star))}{T-\tau^\star}}. \tag{19}$$

In particular, choosing $\tau^\star \asymp L^2T/(L^2+\nu)$ (so that $L^2(T-\tau^\star) \asymp \nu\tau^\star \asymp T$ and $\ell(\tau^\star)$ becomes an absolute constant) replaces the $\sqrt{\log(1+L^2T/\lambda)}$ factor in (18) by an absolute constant, improving $\widetilde{C}$ by a $\sqrt{\log T}$ factor relative to the trivial choice. We do not claim this is the global maximum rigorously; we only use it to compute the leading-order constant. The $\Omega(T)$ bound itself follows from part (a).

Step 3: Phase transition, part (ii). We now read off the regime in which $\sup_\tau \Phi(\tau,T) = \Omega(T)$. Use $\beta_T = O(\sigma\sqrt{r\log T} + \sqrt{\lambda}S)$ and treat $\sigma, \lambda, L, S, \varepsilon_{\mathrm{tail}}$ as positive constants.

(Sufficiency.) By (18) from Step 2, $\sup_\tau \Phi(\tau,T) \ge T\varepsilon_{\mathrm{tail}} - O(r\sqrt{T}\log T)$, which is $\Omega(T)$ iff $r\sqrt{T}\log T = o(T)$, equivalently $r = o(\sqrt{T}/\log T)$.

(Necessity.) Conversely, suppose $\Phi(\tau,T) \ge 0$ for some admissible τ. Squaring the rearrangement $(T-\tau)\varepsilon_{\mathrm{tail}} \ge C\beta_T\sqrt{r(T-\tau)\,\ell(\tau)}$ gives

$$T - \tau \;\ge\; \frac{C^2\beta_T^2\, r\,\ell(\tau)}{\varepsilon_{\mathrm{tail}}^2} \;=\; \Omega\big(r^2\log T\cdot\ell(\tau)\big).$$

For any $\tau \le (1-\eta)T$ with fixed $\eta \in (0,1)$, the logarithm $\ell(\tau)$ is bounded below by a positive constant ($\ell(\tau) \ge \log\big(1 + L^2\eta T/(\lambda+\nu T)\big) = \Omega(1)$), so $T-\tau \ge \Omega(r^2 \log T)$, equivalently $T \ge \Omega(r^2 \log T)$, forcing $r = O(\sqrt{T/\log T})$. The remaining sliver $\tau \in ((1-\eta)T, T-1]$ contributes only a vanishing share of the linear gain $(T-\tau)\varepsilon_{\text{tail}} \le \eta T \varepsilon_{\text{tail}}$, which is $o(T)$ as $\eta \downarrow 0$, so cannot supply $\Omega(T)$. Hence if $r = \omega(\sqrt{T/\log T})$, no τ achieves $\Phi(\tau, T) = \Omega(T)$.

Thus $\sup_\tau \Phi(\tau, T) = \Omega(T)$ iff $r = o(\sqrt{T/\log T})$, and the bound becomes vacuous in the complementary regime $r = \omega(\sqrt{T/\log T})$. The sufficient and necessary directions match at the scaling $r \asymp \sqrt{T/\log T}$ up to a $\sqrt{\log T}$ factor that can be closed by working at the boundary $\tau^\star \asymp L^2 T/(L^2+\nu)$ from Step 2(b). This establishes part (ii) and completes the proof of Theorem 2. □

B.10 Proof of Corollary 2

Proof. We verify the three regimes in turn, using Theorem 2(i) at any convenient τ in each case; appealing to the optimal balance (18) would give the same conclusion up to constants.

Regime (1): $r = o(\sqrt{T/\log T})$. Choose any $\tau = \tau(T)$ with $\tau(T) = o(T)$. Then $T - \tau \sim T$ and $\log\big(1 + L^2(T-\tau)/(\lambda+\nu\tau)\big) = O(\log T)$ uniformly. By Theorem 2(i),

$$T(\bar{r}_T^{\text{meta}} - \bar{r}_T^{\text{fix}}) \ge (T-\tau)\varepsilon_{\text{tail}} - O\big(\beta_T \sqrt{rT\log T}\big).$$

Since $\beta_T = O(\sqrt{r \log T})$ and $r \log T = o(\sqrt{T})$, the second term is $o(T)$, so dividing by T:

$$\bar{r}_T^{\text{meta}} - \bar{r}_T^{\text{fix}} \ge \varepsilon_{\text{tail}} - o(1).$$

This proves the first display in Regime (1); the multiplication factor follows by dividing by $\bar{r}_T^{\text{fix}}$ which is bounded away from 0.

Regime (2): $r = \Theta(\sqrt{T/\log T})$. With $\rho = \lim r \log T/\sqrt{T}$, the same calculation yields

$$\bar{r}_T^{\text{meta}} - \bar{r}_T^{\text{fix}} \ge \varepsilon_{\text{tail}} - C'\rho - o(1)$$

for an explicit $C' > 0$, where C' absorbs the constants in $\beta_T = O(\sqrt{r\log T})$ and the elliptical potential bound. The multiplication conclusion holds whenever $C'\rho < \varepsilon_{\text{tail}}$.

Regime (3): $r = \omega(\sqrt{T/\log T})$. Apply Theorem 2(ii): the right-hand side of (18) is negative for all sufficiently large T because $\beta_T\sqrt{rT\log T} = \omega(T\varepsilon_{\text{tail}})$. Hence $\sup_\tau \Phi(\tau, T) < 0$ asymptotically and Theorem 2(i) provides no nontrivial guarantee. □

C Confidence-Gated Meta-Algorithm: Full Description

This appendix expands the brief description of Section 3.2 into the full per-round walkthrough, states the deterministic confidence-gated routing rule, and provides the pseudocode used in our experiments. The setup recontextualizes the deterministic confidence-gated routing rule of [12]: the regret guarantee in Section 3.2 is conditional on the joint corner condition of Corollary 1.

Rather than encoding prior knowledge through a single improved initialization, we assume access to *multiple frozen policy estimators* $\{(\widehat{\theta}_i, V^{(i)})\}_{i=1}^m$. Each pair should be thought of as a constant policy obtained offline: $\widehat{\theta}_i$ is a fixed linear predictor and $V^{(i)}$ is the corresponding design matrix that captures the geometry of the data on which that predictor was trained. Our goal is not to compete with the single best expert in hindsight, an objective closer in spirit to online model-selection methods such as CORRAL [6, 12], which provide robustness guarantees of the form "perform nearly as well as the best base algorithm." Here we want to exploit reward multiplication *when the corner condition holds* while preserving optimal asymptotic regret in the much larger LinUCB-suffices regime where it does not.

Setting. At each round $t = 1, \dots, T$, the learner observes a finite action set $\mathcal{A}_t$ with feature vectors $\{x_{t,a}\}_{a\in\mathcal{A}_t} \subseteq \mathbb{R}^d$, selects an action $A_t \in \mathcal{A}_t$, and observes reward $r_t = \langle \theta^\star, x_{t,A_t}\rangle + \eta_t$, with η_t conditionally mean-zero sub-Gaussian. We are given m frozen estimators $\{(\widehat{\theta}_i, V^{(i)})\}_{i=1}^m$ and an online LinUCB learner with state $(V_t, b_t, \widehat{\theta}_t)$, where $\widehat{\theta}_t = V_t^{-1} b_t$. The frozen estimators are never updated; only LinUCB is updated online.

Expert proposals and geometry-aware confidence. Each frozen expert $i \in [m]$ proposes the greedy action $a_t^{(i)} \in \arg\max_{a \in \mathcal{A}_t} \langle \widehat{\theta}_i, x_{t,a} \rangle$. To measure how trustworthy expert i is on the current round, we evaluate the uncertainty of its proposed action under its own design geometry,

$$u_t^{(i)} \;:=\; \big\| x_{t,a_t^{(i)}} \big\|_{(V^{(i)})^{-1}}, \qquad \|x\|_{M^{-1}} \;:=\; \sqrt{x^\top M^{-1} x}.$$

The interpretation is simple. If the proposed action lies in directions that are well covered by the offline data underlying expert i, then $u_t^{(i)}$ is small and the expert is more trustworthy; if the current context points in directions poorly represented in $V^{(i)}$, then $u_t^{(i)}$ is large and the expert should be trusted less. We treat LinUCB itself as an additional candidate, indexed by 0, with proposal $a_t^{(0)} \in \arg\max_a \langle \widehat{\theta}_t, x_{t,a} \rangle + \alpha_t \|x_{t,a}\|_{V_t^{-1}}$ and uncertainty score $u_t^{(0)} := \|x_{t,a_t^{(0)}}\|_{V_t^{-1}}$.

Deterministic confidence-gated selection. The natural rule is to prefer the frozen expert whose proposed action has the smallest uncertainty score, unless the online learner is already at least as confident. Concretely, with $i_t^\star \in \arg\min_{i \in [m]} u_t^{(i)}$,

$$I_t \;=\; \begin{cases} i_t^\star, & \text{if } u_t^{(i_t^\star)} < u_t^{(0)}, \\ 0, & \text{otherwise,} \end{cases} \qquad A_t \;:=\; a_t^{(I_t)}.$$

The master is entirely deterministic. Early in learning, LinUCB typically has a large confidence radius, so a frozen expert may be preferred; as data accumulate, V_t grows and the LinUCB uncertainty shrinks, and eventually the adaptive learner becomes the most confident option and takes over automatically.

Optional permanent handover to LinUCB. To formalize the idea that LinUCB should eventually dominate, we include a one-way switching criterion. With upper-confidence scores $U_t^{(i)} = \langle \widehat{\theta}_i, x_{t,a_t^{(i)}} \rangle + \alpha_i \|x_{t,a_t^{(i)}}\|_{(V^{(i)})^{-1}}$ for each frozen expert and lower-confidence score $L_t^{(0)} = \langle \widehat{\theta}_t, x_{t,a_t^{(0)}} \rangle - \alpha_t \|x_{t,a_t^{(0)}}\|_{V_t^{-1}}$ for LinUCB, if $L_t^{(0)} \geq \max_{i \in [m]} U_t^{(i)}$ for K consecutive rounds we permanently hand control to LinUCB.

Distinction from randomized model selection. The deterministic, sample-by-sample geometric routing distinguishes Algorithm 1 from CORRAL-style randomized masters in two ways. First, it can exploit per-context information about which estimator is reliable on *this particular* action set; second, because it is deterministic, the LinUCB design matrix receives *passive covariance updates* from the expert's actions, transferring geometric information directly into the adaptive learner's state. This is the mechanism that drives the multiplicative shrinkage factor in Theorem 1; randomized mixing does not provide this transfer.

D Pre-Runtime Diagnostics, Deployment Rule, and LLM-Quality Bars

This appendix consolidates two questions that the body distinguishes but that are easy to conflate. The first question is *whether to deploy the meta-algorithm at all* on a given instance; the second is *whether, once deployed, it realizes its theoretical multiplicative payoff*.

Two scopes. The deployment question is answered by four pre-runtime diagnostics computed on an offline sample of contexts: three of them (effective rank, coherence, oracle gap distribution) depend only on the instance, and a fourth (subspace mismatch) is an LLM-fitness check folded into the same instance-level classifier. The classifier outputs one of {LinUCB-suffices, gap-suffices, warm-start-essential, ambiguous}, and the deployment rule reads off this output without ever inspecting the LLM. The payoff question is answered separately by three LLM-quality conditions Q1–Q3, which restate Assumptions A, B, and D of Section 3.2 in LLM-facing language and pair each with a held-out proxy. Q1–Q3 govern *whether the deployed meta-algorithm earns its theoretical multiplier*; they do not govern *whether to deploy*. The structure of this appendix mirrors that distinction: Subsections D.1–D.3 cover the deployment story (instance-only); the transition paragraph at the end of Subsection D.3 explicitly hands off to Subsections D.4–D.6, which cover the realized-payoff story (LLM-conditional).

Algorithm 1 Confidence-Gated Master over Frozen Policies and LinUCB

Require: Frozen estimators $\{(\widehat{\theta}_i, V^{(i)})\}_{i=1}^m$, regularization $\lambda > 0$, confidence parameters $\{\alpha_t\}_{t=1}^T$

Initialize LinUCB state:

$$V_1 = \lambda I_d, \qquad b_1 = 0, \qquad \widehat{\theta}_1 = 0$$

Set HANDOVER $\leftarrow$ False
for $t = 1, \ldots, T$ **do**
  Observe action set $\mathcal{A}_t$
  Compute LinUCB proposal

$$a_t^{(0)} \in \arg\max_{a \in \mathcal{A}_t} \left\{ \langle \widehat{\theta}_t, x_{t,a} \rangle + \alpha_t \|x_{t,a}\|_{V_t^{-1}} \right\}$$

  Set $u_t^{(0)} \leftarrow \|x_{t,a_t^{(0)}}\|_{V_t^{-1}}$
  for $i = 1, \ldots, m$ **do**
    Compute frozen proposal

$$a_t^{(i)} \in \arg\max_{a \in \mathcal{A}_t} \langle \widehat{\theta}_i, x_{t,a} \rangle$$

    Set $u_t^{(i)} \leftarrow \|x_{t,a_t^{(i)}}\|_{(V^{(i)})^{-1}}$
  end for
  if HANDOVER $=$ True **then**
    $A_t \leftarrow a_t^{(0)}$
  else
    Choose $i_t^\star \in \arg\min_{i \in [m]} u_t^{(i)}$
    if $u_t^{(i_t^\star)} < u_t^{(0)}$ **then**
      $A_t \leftarrow a_t^{(i_t^\star)}$
    else
      $A_t \leftarrow a_t^{(0)}$
    end if
  end if
  Observe reward r_t
  Update LinUCB:

$$V_{t+1} \leftarrow V_t + x_{t,A_t} x_{t,A_t}^\top, \qquad b_{t+1} \leftarrow b_t + r_t x_{t,A_t}, \qquad \widehat{\theta}_{t+1} \leftarrow V_{t+1}^{-1} b_{t+1}$$

  Compute $U_t^{(i)}$ for all $i \in [m]$ and $L_t^{(0)}$
  if $L_t^{(0)} \geq \max_{i \in [m]} U_t^{(i)}$ for K consecutive rounds **then**
    HANDOVER $\leftarrow$ True
  end if
end for

D.1 Formal Definitions of the Four Diagnostics

Throughout, fix an offline sample of n contexts $\{x_i\}_{i=1}^n \subset \mathbb{R}^d$ drawn i.i.d. from the context distribution and an action embedding map $a \mapsto z_a \in \mathbb{R}^d$. Let $\widehat{\Sigma} := \frac{1}{n} \sum_{i=1}^n x_i x_i^\top$ denote the empirical context covariance.

(A) Effective rank. $\widehat{r} := \mathrm{tr}(\widehat{\Sigma}) / \|\widehat{\Sigma}\|_{\mathrm{op}}$. This is a stable real-valued surrogate for the matrix rank: for any $\widehat{\Sigma} \succeq 0$, $1 \leq \widehat{r} \leq \mathrm{rank}(\widehat{\Sigma})$, with equality on the right when the nonzero eigenvalues of $\widehat{\Sigma}$ are all equal. Operationally, $\widehat{r}/\sqrt{T} < 1$ places the instance in the LinUCB-suffices regime on axis A; this is a slightly looser check than the asymptotic condition $\widehat{r} \log T \leq c\sqrt{T}$ of Lemma 1 since the $\log T$ factor is absorbed into the constant c.

(B) Coherence. $\widehat{\bar{\alpha}} := \mathrm{median}_{a \in \mathcal{A}} \min_{a' \neq a} \cos \angle(z_a, z_{a'})$, the median over a of the minimum pairwise cosine similarity between z_a and any other action embedding. The median is robust to

outlier actions; the inner minimum captures the worst-case alignment seen by action a. Instances with $\widehat{\alpha} \geq \alpha_0$ for a strictly positive threshold $\alpha_0 > 0$ (we use $\alpha_0 = 0.1$, matching the body) lie in the LinUCB-suffices regime on axis B.

(C) Gap distribution. Fit an oracle linear model $\widehat{\theta}_{\text{oracle}}$ by ridge regression on the joint features $\phi(x, a) := x \odot z_a$ using ground-truth rewards. For each held-out context x, compute $\widehat{\Delta}(x) := \widehat{\theta}_{\text{oracle}}^{\top}\phi(x, a^{(1)}) - \widehat{\theta}_{\text{oracle}}^{\top}\phi(x, a^{(2)})$, the margin between the oracle's top-1 and top-2 predictions. The empirical gap distribution is heavy at zero if $|\{x : \widehat{\Delta}(x) \leq 0.01\}|/n \geq 0.5$; this indicates Lemma 7, otherwise Lemma 3 applies.

(A+C) Subspace mismatch. Let $\Pi_r^{(i^\star)}$ be the orthogonal projector onto the top-r eigenvectors of $V^{(i^\star)}$ (the design matrix of the best frozen expert as scored by the diagnostic) and $\widehat{\Pi}_r$ the analogous projector for $\widehat{\Sigma}$, with $r = \widehat{r}$. The subspace mismatch is the maximum principal angle $\angle_{\max}(\Pi_r^{(i^\star)}, \widehat{\Pi}_r)$. Large maximum angles indicate that at least one direction in the online context covariance is orthogonal to the warm-start cover, so the warm-start cannot contract the LinUCB confidence ellipsoid uniformly over the reward-relevant subspace. Note that this is the only diagnostic that references a candidate frozen expert; we use it as an LLM-fitness check inside the same instance-level classifier rather than as a separate condition.

D.2 Consistency Lemmas

The diagnostics are useful only insofar as the empirical statistics they compute concentrate around their population analogues. We record two standard concentration results in our notation.

Lemma 10 (Effective-rank concentration)**.** *Suppose contexts $x_i \in \mathbb{R}^d$ have $\|x_i\|_2 \leq L$ a.s. and covariance $\Sigma := \mathbb{E}[xx^\top]$ with effective rank $r := \operatorname{tr}(\Sigma)/\|\Sigma\|_{\text{op}}$. With probability at least $1 - \delta$ over the sample,*

$$\left|\widehat{r} - r\right| \;\leq\; \frac{C\, r\, L^2}{\|\Sigma\|_{\text{op}}}\sqrt{\frac{\log(d/\delta)}{n}}$$

for a universal constant C. In particular, with $n \geq \Omega(r^2 L^4 \log(d/\delta))$ samples and provided $\|\Sigma\|_{\text{op}} = \Theta(1)$, the diagnostic $\widehat{r}$ is within a constant factor of the population effective rank r with probability $1 - \delta$.

Proof sketch. Apply the matrix Bernstein inequality (Theorem 6.6.1 of [27]) to bound $\|\widehat{\Sigma} - \Sigma\|_{\text{op}}$ by $O(L^2\sqrt{\log(d/\delta)/n})$. The trace error is bounded *directly* via scalar concentration on the bounded random variable $\|x\|_2^2 \leq L^2$: by Hoeffding's inequality, $|\operatorname{tr}(\widehat{\Sigma}) - \operatorname{tr}(\Sigma)| = \left|\frac{1}{n}\sum_i \|x_i\|_2^2 - \mathbb{E}\|x\|_2^2\right| \leq L^2\sqrt{\log(2/\delta)/n}$ with probability $1 - \delta/2$, avoiding any spurious dependence on d. The operator norm continuity of $1/\|\cdot\|_{\text{op}}$ holds under $\widehat{\Sigma} \succeq \Sigma/2$, which follows from the same matrix Bernstein bound with probability $1 - \delta/2$. Combining gives the stated bound. □

Lemma 11 (Gap-distribution concentration)**.** *Suppose contexts x_i are i.i.d. and the gap function $\Delta : \mathcal{X} \to [0, 1]$ is bounded. Let $F(\eta) := \Pr[\Delta(x) \leq \eta]$ denote the population gap CDF and $\widehat{F}(\eta)$ its empirical analogue. Then by the Dvoretzky–Kiefer–Wolfowitz inequality, with probability at least $1 - \delta$,*

$$\sup_{\eta}\left|\widehat{F}(\eta) - F(\eta)\right| \;\leq\; \sqrt{\log(2/\delta)/(2n)}.$$

In particular, the heavy-left-tail diagnostic $\widehat{F}(0.01) \geq 0.5$ correctly indicates the population condition $F(0.01) \geq 0.5 - \sqrt{\log(2/\delta)/(2n)}$ uniformly.

The two consistency lemmas above are the formal counterparts of the operational claim that "the diagnostics are predictive of the regime" made in Section 3.1. Concentration of $\widehat{\alpha}$ and the principal-angle diagnostic follows by analogous arguments (matrix Bernstein for $\widehat{\Sigma}$ and Davis–Kahan for the projector $\widehat{\Pi}_r$, respectively); we omit the details for brevity.

D.3 Diagnostic-Driven Deployment Rule

The four diagnostics give a pre-runtime classifier $\mathsf{Regime}(\mathcal{D}) \in \{\textsf{LinUCB-suffices}, \textsf{gap-suffices}, \textsf{warm-start-essential}, \textsf{ambiguous}\}$ according to the decision rule:

- if $\widehat{r}/\sqrt{T} < 1$ *or* $\widehat{\tilde{\alpha}} \geq \alpha_0$ *or* the subspace mismatch exceeds a threshold $\angle_0$ (we use $\alpha_0 = 0.1$ and $\angle_0 = 45°$, matching the body), output LinUCB-suffices;
- else if $\widehat{F}(0.01) < 0.5$ (no heavy gap-tail) *and* $\widehat{r}$ is moderate enough that $\widehat{r}^{\,2} \log^2 T/\Delta_{\min} = o(T)$ (so that the gap-dependent bound of Lemma 3 is non-vacuous), output gap-suffices;
- else if all of $\widehat{r}/\sqrt{T} \geq 1$, $\widehat{\tilde{\alpha}} < \alpha_0$, the subspace mismatch is below $\angle_0$, and $\widehat{F}(0.01) \geq 0.5$, output warm-start-essential;
- otherwise output ambiguous.

This classifier is the deployment rule used in the experiments of Section 4: run cold-start LinUCB on LinUCB-suffices and gap-suffices instances; run the meta-algorithm of Appendix C on warm-start-essential instances; track both in parallel on ambiguous instances. The deployment decision is purely instance-level: the meta-algorithm is a tool tuned for one specific tree leaf, and the diagnostics tell the practitioner whether they reach that leaf.

Bridge to the realized-payoff story. The classifier above answers *whether to deploy the meta-algorithm*, and that is the only question on which the deployment decision depends. A separate question is whether, on instances where the meta-algorithm is deployed, it actually realizes the multiplicative payoff guaranteed by Theorem 2. That payoff depends on properties of the LLM–instance pair that no instance-level statistic can certify on its own — in particular, on the LLM being accurate enough on expert-routed rounds, on its routed actions covering the reward-relevant subspace, and on its tail accuracy being strictly below 1. The remainder of this appendix (Subsections D.4–D.6) formalizes these as conditions Q1–Q3 and pairs each with a held-out proxy. Crucially, Q1–Q3 do not enter the deployment rule.

D.4 LLM-Quality Bars Q1–Q3

This subsection introduces the three LLM-quality conditions that govern whether the meta-algorithm's deployed multiplier is realized. Each condition is the LLM-facing restatement of one of the assumptions of Theorem 1; the equivalence is recorded explicitly below and is used in the proof of Proposition 2. *Throughout this and the following two subsections, we assume the instance has been classified as warm-start-essential by the deployment rule of Subsection D.3.* Q1–Q3 are conditions on the LLM–instance pair given that classification, not conditions that themselves classify the instance.

Recall the random partition $\mathcal{T}_{\mathrm{exp}} \sqcup \mathcal{T}_{\mathrm{lin}}$ of rounds into expert-routed and LinUCB-routed (with $\tau := |\mathcal{T}_{\mathrm{exp}}|$), and the instantaneous gap of the routed expert $\Delta_t^{(I_t)} := \mu_t(a_t^\star) - \mu_t(a_t^{(I_t)})$.

(Q1) Accuracy. There exists $\varepsilon_0 \geq 0$ with $\Delta_t^{(I_t)} \leq \varepsilon_0$ for every $t \in \mathcal{T}_{\mathrm{exp}}$, and $\varepsilon_0 \ll \beta_T \sqrt{r/T}$. The LLM is uniformly near-optimal on the rounds where it is consulted; the explicit inequality on ε_0 ensures the warm-start phase does not accumulate $\Theta(R_T^{\mathrm{cold}})$ regret on its own. This is exactly Assumption A of Section 3.2.

(Q2) Coverage. There exists $\nu > 0$ such that

$$\sum_{t \in \mathcal{T}_{\mathrm{exp}}} x_{t,A_t} x_{t,A_t}^\top \;\succeq\; \nu\tau\, P,$$

where P is the projector onto the reward-relevant subspace. The LLM-routed actions span this subspace at rate ν per round. This is exactly Assumption B of Section 3.2.

(Q3) Eventual suboptimality. There exists $\varepsilon_{\mathrm{tail}} > 0$ with

$$\frac{1}{T-\tau} \sum_{t=\tau+1}^{T} \left(\mu_t(a_t^\star) - \mu_t(a_t^{(i)})\right) \;\geq\; \varepsilon_{\mathrm{tail}}$$

for every frozen expert $i \in [m]$. This is Assumption D of Section 3.2; without it the LLM is already near-optimal on the post-handover horizon and there is no linear payoff to harvest.

D.5 Dataset-Measurable Proxies for Q1–Q3

Q1, Q2, Q3 are properties of the LLM–instance pair; they cannot be read directly from offline statistics, but each admits a measurable proxy.

Lemma 12 (Q1 proxy: held-out top-1 accuracy)**.** *Let $\widehat{\mathrm{acc}}_n$ denote the empirical top-1 accuracy of the LLM-derived policy on a held-out sample of size n. Under the linear reward model with $\|\theta^\star\|_2 \le S$ and bounded features,*

$$\mathbb{E}[\Delta_t^{(I_t)}] \;\le\; 2LS(1-\widehat{\mathrm{acc}}_n) \;+\; O\big(\sqrt{\log(1/\delta)/n}\big)$$

with probability $1-\delta$ over the sample. Hence $\widehat{\mathrm{acc}}_n \ge 1-\varepsilon_0/(2LS)$ certifies Q1 in expectation up to $O(\sqrt{\log(1/\delta)/n})$ corrections.

The accuracy bound translates directly: an LLM with 90% held-out accuracy on a problem with $L = S = 1$ certifies $\varepsilon_0 \le 0.2$ in expectation. The bound is loose but operationally useful: it gives a single number practitioners can compute before deployment.

Lemma 13 (Q2 proxy: subspace-mismatch diagnostic)**.** *Let $\angle_{\max}$ denote the* maximum *principal angle between the rank-r subspaces of $\mathbb{E}[\sum_{t\in\mathcal{T}_{\exp}} x_{t,A_t}x_{t,A_t}^\top]$ and P (the reward-relevant projector). Then*

$$\sum_{t\in\mathcal{T}_{\exp}} \mathbb{E}[x_{t,A_t}x_{t,A_t}^\top] \;\succeq\; \big(\cos^2\angle_{\max}\big)\nu_0\tau\, P$$

for $\nu_0 := \lambda_{\min}(\mathbb{E}[xx^\top | x \in \mathrm{Im}(P)])$, the minimum eigenvalue of the context covariance restricted to the reward-relevant subspace. The Loewner bound is governed by the worst *(largest) principal angle, since the smallest eigenvalue of the projection onto P is determined by whichever direction in $\mathrm{Im}(P)$ is least aligned with the expert subspace. The subspace-mismatch diagnostic of Subsection D.1 is therefore monotone in the implied coverage rate $\nu = \nu_0\cos^2\angle_{\max}$: small maximum mismatch implies high ν, hence Q2. Note that this is the same statistic used as an LLM-fitness check inside the deployment classifier of Subsection D.3; it appears in both scopes because it is computable from the instance* and *informative about the LLM's coverage.*

Lemma 14 (Q3 proxy: held-out tail accuracy)**.** *Let $\widehat{\mathrm{acc}}_n^{\,\mathrm{tail}}$ denote the empirical top-1 accuracy of the LLM-derived policy on a held-out sample drawn from the post-handover round distribution (i.e., the test split representing rounds $t > \tau$). Then*

$$\frac{1}{T-\tau}\sum_{t=\tau+1}^{T}\mathbb{E}[\mu_t(a_t^\star) - \mu_t(a_t^{(i)})] \;\ge\; (1-\widehat{\mathrm{acc}}_n^{\,\mathrm{tail}})\,\Delta_{\min}^{\mathrm{tail}} \;-\; O\big(\sqrt{\log(1/\delta)/n}\big),$$

where $\Delta_{\min}^{\mathrm{tail}}$ is the minimum gap on post-handover rounds, with probability $1-\delta$. Hence $\widehat{\mathrm{acc}}_n^{\,\mathrm{tail}} < 1$ together with a positive minimum gap certifies Q3.

D.6 The Compound Multiplication Condition

Multiplication requires Q1, Q2, Q3 *and* the warm-start-essential regime simultaneously — the first three coming from the LLM's quality on the instance, the last from the structural diagnostics of Subsection D.3. The dimensional threshold $r = o(\sqrt{T}/\log T)$ from Theorem 2 (axis A in disguise) determines whether the linear payoff $T\varepsilon_{\mathrm{tail}}$ from Q3 dominates the $\widetilde{O}(\beta_T\sqrt{rT})$ post-handover learning cost; together with Q1 and Q2 it yields the "LLM is good enough *and* the problem is structured enough" compound condition for genuine multiplication.

Proposition 2 (Compound multiplication condition)**.** *Assume Q1–Q3 hold with parameters $(\varepsilon_0, \nu, \varepsilon_{\mathrm{tail}})$, the regime is warm-start-essential per the deployment rule of Subsection D.3, and $r = o(\sqrt{T}/\log T)$. Then the meta-algorithm of Appendix C satisfies*

$$\liminf_{T\to\infty}\frac{\bar r_T^{\mathrm{meta}}}{\bar r_T^{\mathrm{fix}}} \;\ge\; 1 + \frac{\varepsilon_{\mathrm{tail}}}{\bar r_\infty^{\mathrm{fix}}}.$$

If any one *of Q1–Q3 or the warm-start-essential condition fails, no multiplicative improvement strictly above $1 + o(1)$ is guaranteed by Theorem 2.*

Proof. The first claim is exactly Corollary 2(1) under Assumptions A, B, and D, which are equivalent to Q1–Q3 (Q1≡Assumption A, Q2≡Assumption B, Q3≡Assumption D); the linear reward model of Theorem 1 is part of the setup. The second claim is the contrapositive: if any of Q1–Q3 fails, the corresponding term in the regret bound becomes trivial; if the warm-start-essential condition fails, Cor. 1 guarantees only a constant-factor improvement over cold-start LinUCB. □

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